

Physical Vapor Deposition

Yoke Khin Yap and Dongyan Zhang
Department of Physics, Michigan Technological
University, Houghton, MI, USA

Synonyms

Anodic arc deposition; Arc discharge; Boron nitride nanotubes (BNNTs); Carbon nanotubes (CNTs); Catalyst; Cathodic arc deposition; Chemical beam epitaxial (CBE); Electron beam evaporation; Electron beam physical vapor deposition (EBPVD); Gallium arsenide (GaAs); Gas-phase molecular beam epitaxy (gas-phase MBE); Graphene; Ion beam; Magnetron sputtering; Metal-organic molecular beam epitaxy (MOMBE); Molecular beam epitaxy (MBE); Nanorods; Nanotubes; Nanowires; Plasma; Pulsed laser deposition (PLD); Si nanotubes; Sputtering; Thermal evaporation

Definition

Physical vapor deposition (PVD) is referred to deposition processes of thin films and nanostructures through the evaporation of solid precursors into their vapor phase by physical approaches followed by the condensation of the vapor phase on substrates. The whole process consists of three stages: (1) evaporation of the

solid source, (2) vapor phase transport from the source to the substrates, and (3) vapor condensation on the substrates. Since PVD technique can convert solid materials into vapor phase without chemical processes, they are very convenient and possible to deposit many types of materials into thin films and nanostructures.

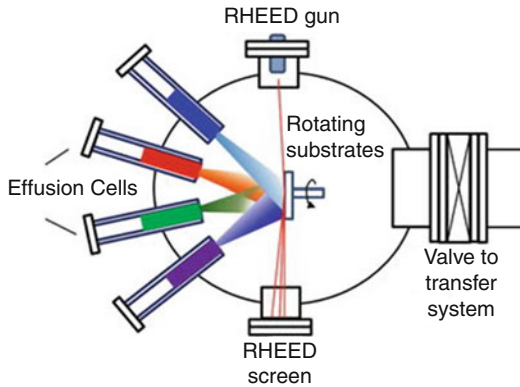
Classification

PVD techniques can be classified based on the techniques used to evaporate the solid source materials into their vapor phase [1, 2]. These techniques include (1) thermal evaporation, (2) ion sputtering, and (3) arc discharge. In the following subsections, examples of PVD based on these evaporation techniques will be described.

PVD by Thermal Evaporation

Molecular Beam Epitaxy

Molecular beam epitaxy (MBE) is a PVD technique conducted under ultrahigh vacuum (UHV, better than 10^{-9} mbar) where epitaxial deposition of thin film crystal on crystalline substrates can be obtained by using one or more vapor sources (*effusion cells* or sometimes called *Knudsen cells*). An effusion cell consists of a crucible (made of graphite, pyrolytic boron nitride, quartz, or tungsten) where the solid source can be heated and evaporated by hot filaments (Fig. 1). For



Physical Vapor Deposition, Fig. 1 Schematic drawing of a typical molecular beam epitaxy system

example, powders of gallium (Ga) and arsenic (As) can be evaporated in separated cells until they are sublimate into vapors. When these separated beams of element vapors are condensed on a heated Si substrate, crystallized thin films of gallium arsenide (GaAs) will be formed. A typical MBE system has a reflection high-energy electron diffraction (RHEED) system to provide in situ monitoring of the crystal growth processes.

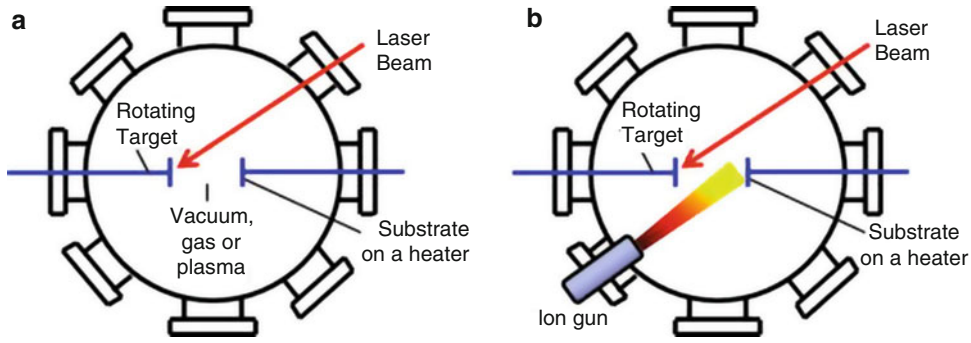
The major advantage of MBE comes from its controllable deposition rate at the atomic or molecular scale. This can be achieved due to the use of effusion cells that enable the generation of “molecular beam,” where the generated vapors have low interparticle collisions before they reach to the substrate surface. This means the evaporated atoms escape from the orifice of the cells through effusion and have long mean free path. They neither interact with each other nor with the gases in the vacuum chamber. Due to this capability, MBE was recently used for the growth of graphene, but the outcome is debatable [3, 4]. Finally, gases were also used as the precursors instead of solid sources, and such configuration is called *gas-phase MBE* or more appropriately *chemical beam epitaxial (CBE)* [5]. In this case, the pyrolysis of either trimethylindium (TMIn) or triethylindium (TEIn) and trimethylgallium (TMGa) or triethylgallium (TEGa) at the heated substrates was used as the In and Ga source, while As_2 and P_2 were obtained by thermal

decomposition of the trimethylarsine (TMAs) and triethylphosphine (TEP). Such approach is more like a combined MBE and *metal-organic chemical vapor deposition (MOCVD)* and is sometimes called *metal-organic molecular beam epitaxy (MOMBE)*.

In fact, *MBE* is an advanced form of thermal evaporation technique with highly controllable deposition thickness (and thus with the drawback of slow deposition rate) due to the use of effusion cells and high vacuum. For the deposition of amorphous thin films, much simple setup can be employed with less demanding vacuum level ($\sim 10^{-2}$ mbar in a glass bell jar chamber). This approach is simply called *direct resistive evaporation* and can be achieved by passing a large current through a resistive wire or foil containing the solid material to be deposited. The heating element is often referred to as an “evaporation source.” The evaporation source can appear in many forms including tungsten wires in the form of filaments or baskets and thin tungsten, tantalum, or molybdenum foils in the form of boats.

PVD by Electron Beam (E-Beam) Evaporation

In contrast to *MBE*, *electron beam physical vapor deposition (EBPVD)* provides very high deposition rate on large deposition areas. *EBPVD* is referring to thin film deposition initiated by electron beam evaporation. The deposition rate is typically ranging from tens of nanometer/minute to hundreds of micrometer/minute, although a rate as high as $50 \mu\text{m/s}$ was reported [6]. *EBPVD* is based on electron beam evaporation of anode materials (usually in ingot form) in high vacuum (10^{-4} mbar). A typical *EBPVD* system can have multiple electron guns and sources. The electron beams can be generated by either thermionic emission or field emission of filaments. These beams are then accelerated to a high kinetic energy and focused on the ingot. The energy of these electrons is converted into thermal energy as the beam bombards the surface of the ingot. The high deposition rate of *EBPVD* without corrosive products meets the needs of industrial applications. A potential drawback of *EBPVD* is that filament degradation in the



Physical Vapor Deposition, Fig. 2 Schematic drawings of a typical pulsed laser deposition (PLD) system operated (a) in vacuum or with auxiliary plasma or (b) with auxiliary ion beam

electron gun results in a nonuniform evaporation rate. In addition, *EBPVD* may cost contamination of the deposition chamber as the vapors generated from the source materials are coated everywhere in the vacuum chamber, the so-called line-of-sight process.

Pulsed Laser Deposition

Pulsed laser deposition (PLD) is a *PVD* technique where a high-power pulsed laser is used to generate vapors (i.e., laser ablation) from a solid target followed by the condensation of the vapors on substrates. The experimental setup for *PLD* is relatively simpler than many *PVD* techniques, but the laser ablation mechanism is quite complex [7]. As shown in Fig. 2a, a *PLD* system consists of a pulsed laser, a vacuum chamber with a rotating target holder, and a substrate holder with a heater. *PLD* can be performed in vacuum at various substrate temperatures. During film deposition, laser pulses are focused on the surface of a solid target through a viewport of the chamber by using a thin lens. This will generate a laser spot with very high *irradiance* (I , power per unit area) and local electric field, where $I = \frac{1}{2}\epsilon_0 c E_o^2$ [8]. Here, E_o is the amplitude of electric field of the laser, c is the speed of light in vacuum, and ϵ_0 is the vacuum permittivity. The local electric field is sufficient to cause atomic bond breaking and dielectric breakdown of the target materials to initiate vaporization of the target materials. This vapor will absorb the remaining portion of the laser pulse, and the electric field of the pulse will accelerate electrons and ions inside the vapor

to create rapid heating and to form laser plasma. Continued absorption of the laser pulse will cause a highly directional expansion of the laser plasma perpendicular to the target surface. This expanding vapor is called laser plume that propagates toward the substrate placed several centimeters opposite to the target.

PLD can be conducted in vacuum or in a low-pressured gas ambient. When *PLD* is conducted in reactive gas ambient (say oxygen or nitrogen), the technique is called *reactive PLD*. In this case, the expanding laser plume will form shock waves and cause chemical reaction between the target vapor and the gas molecules. *PLD* can also be modified into various configurations. For example, plasma can be applied to enhance the interaction of the vapors with the gas ambient, and the technique is called *plasma-enhanced or plasma-assisted PLD*. For example, such an approach was shown to control the phases of nitride thin films [9, 10]. Sometimes, an auxiliary ion beam is irradiated on the surface of the substrates during thin film deposition as shown in Fig. 2b, and the technique is called *ion beam-assisted PLD*.

In fact, the mechanism of laser ablation and the quality of samples deposited by *PLD* depend strongly on the wavelength and energy of the pulsed laser. This can be understood by the absorption of the laser light by the laser plasma when the plasma index of refraction, $n(\omega)$, becomes complex. According to the relation $n^2(\omega) = 1 - [\omega_p/\omega]^2$, a complex $n(\omega)$ will occur when $\omega_p > \omega$ [8]. This means, shorter laser

wavelength (larger ω) will reduce the chances of such plasma heating effect. However, one should also consider that the laser plasma frequency, $\omega_p = [N e^2 m_e \epsilon_0]^{1/2}$, where m_e is the mass of electron and N is the electron density. In this case, to avoid plasma heating, N must be minimized, which can be obtained by keeping the laser energy or irradiance low. Plasma heating will lead to the so-called photothermal ablation. For example, photothermal ablation of h-BN target will transfer h-BN target to the substrates and prevent the formation of cubic phase (c-BN) [8]. Photothermal ablation of graphite target at longer laser wavelength will also prevent the formation of diamond-like carbon film [11] as the sp^2 species from the graphite are transferred to the substrates.

PVD by Sputtering

Sputtering is a process where atoms are released from a solid target material due to bombardment of the target by energetic particles such as ions and atoms. The atoms released from the target can then be deposited as thin film on substrates and is commonly referred as PVD by sputtering. The physics behind the process is momentum transfer from the incident ions/atoms to the target materials through collisions. The average number of atoms ejected from the target per incident particle is called the sputter yield. The yield depends on the ion incident angle, the energy of the ion, the masses of the ion and target atoms, the surface binding energy of atoms in the target, and the crystalline orientation of the target when crystals are used as the target. There are many approaches to generate sputter including the use of DC, AC, and RF plasmas or external ion beams.

Sputtering offers several advantages over other PVD techniques including (1) large area deposition as compared to PLD, (2) convenience for thin film deposition of alloy/composite and materials with high melting points, and (3) avoidance of device damaging from X-rays generated by electron beam evaporation. However, sputtering also has the following disadvantages: (1) more complicated setup than PLD, (2) contamination more likely to occur due to the use of

plasma as well as relatively lesser vacuum level, and (3) the film morphology that may be rougher and even damaged due to bombardment of energetic growth species and clustering of the growth species in the relatively high deposition pressures.

PVD by DC, AC, RF, and Magnetron Sputtering

PVD can be obtained by applying plasma in various configurations. The simplest one is to apply a DC potential across the substrate (anode) and target (cathode). In this case, plasma can be generated when gases are introduced (10–100s of mTorr) at sufficient electric field strength ($\sim$ a few kV/cm). For example, argon gas is often used, as argon ion can lead to high sputter yield and uniform glow discharge. These Ar ions will bombard the target surface to generate vapors that will be deposited to the substrates. Despite the simplicity, the DC configuration will have the following issues: (1) positive charges will build up on the target surface and cause sparking within the plasma and (2) not suitable for insulating target. To avoid these issues, AC and RF configurations are often used. In these cases, the potential between the target and substrate is alternating so that the target will be negative in potential more often (to generate sputtering) than positive potential. These configurations will also allow lower operation pressure ($\sim$ 10–30 mTorr).

The use of magnetron can further enhance the deposition rate at even lower pressures by the use of DC, AC, or RF electric fields. RF potential is of advantage, since it can be used for insulating targets and is applicable in an antenna configuration, that is, without positive bias voltages on the substrates (less electron bombardments and less substrate heating). In this case, the target is placed on the magnetron surrounded by strong electric and magnetic fields. Electrons in the RF plasma will follow helical paths around the magnetic field lines and generate more ionizing collisions near the target surface with the surrounding gas. Thus, more ions will be created and will lead to a higher sputtering and deposition rates. Because of higher ionization rates, plasma can be sustained at a lower pressure.

PVD by Ion Beam Sputtering

Ion beam sputtering (*IBS*) is a PVD technique in which the target is evaporated by an external ion source (ion beam). In a typical ion source, ions are generated by collisions of neutral gas atoms/molecules with electrons. They are then accelerated by the electric field by a grid electrode toward the target. Before these ions leave the source, they are neutralized by electrons from a second external filament. The major advantage of PVD by *IBS* is that the energy and flux of ions can be controlled independently. In addition, the *IBS* approach is applicable to insulating and conducting targets since the flux of “ions” (which are neutralized as they are emitted from the ion beam) that strikes the target is composed of neutral atoms/molecules.

Reactive Sputtering

Like the PLD process, reactive sputtering can be achieved by mixing reactive gases (N_2 , O_2 , etc.) with the inert gas (Ar, Kr, etc.). While the inert gas is responsible for the sputtering of the target materials, the reactive gases are used to initiate chemical reactions with the target vapors to form oxide or nitride films. Like *PLD*, reactive sputtering can also be achieved by various configurations. For example, plasma can be applied to enhance the interaction between the target vapors and the reactive gas ambient, and the technique is called *plasma-enhanced or plasma-assisted deposition*. Sometimes, an auxiliary ion beam is irradiated on the surface of the substrates during thin film deposition, and the technique is called *ion beam-assisted deposition (IBAD)*.

PVD by Arc Discharge

In addition to PVD by thermal evaporation and sputtering, the desired vapors for thin film deposition can be obtained by arc discharge of the target. This is sometimes called *arc vapor deposition* [1], where a high-current (10^4 – 10^6 A/cm²), low-voltage (tens of voltages) discharge was applied on the target. For example, *cathodic arc deposition* refers to arc vapor deposition using the target as the cathode. In this case, a high-voltage “trigger arc” will be ignited between the cathode and a sharp auxiliary anode. This will form

“seed” electrons and ions to initiate a low-voltage, high-current discharge between the target cathode and the adjacent anode.

On the other hand, target materials can be evaporated by using the target as the anode of the arc discharge. This PVD process is referred as *anodic arc deposition*. In fact, this process is similar to *electron beam physical vapor deposition (EBPVD)* discussed earlier, where arc discharge is generated by irradiating an unfocused electron beam on the target anode. The electrons can be made to spiral in a magnetic field to further ionize the evaporated materials from the target prior to deposition. Like most other PVD processes, *reactive arc vapor deposition* can be performed by both *anodic arc deposition* and *cathodic arc deposition* using reactive gases like nitrogen and oxygen. This will lead to the formation of nitride and oxide films.

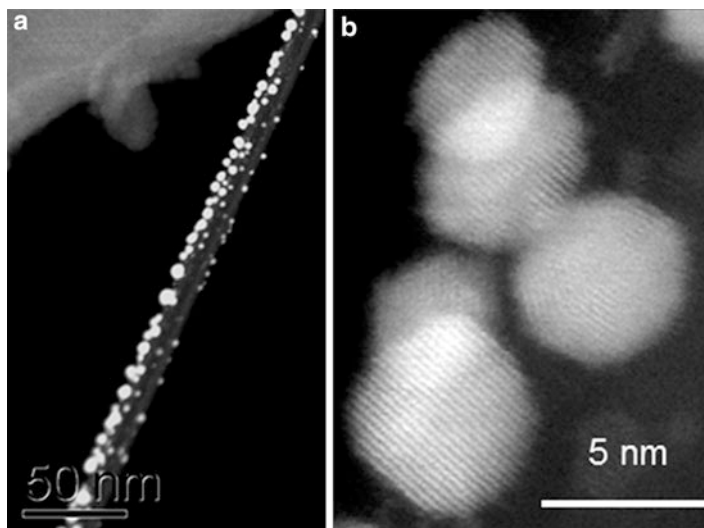
The deposition rates of *arc vapor deposition* are usually higher than that of PVD by sputtering. This is one of the advantages of *anodic arc deposition* and *cathodic arc deposition* in addition to the low operation voltage. On the other hand, contamination is the potential drawback of most plasma-based PVD processes. In addition, particulates or the so-called macros are undesired for thin films deposited by *anodic arc deposition* and *cathodic arc deposition*. Macros are formed by ablation of molten or solid particles by thermal shock during arc discharges.

Examples of PVD Approaches for Nanotechnology

Physical evaporation techniques like arc discharge [12] and laser ablation [13] are important on the discovery of carbon nanotubes (CNTs). However, PVD techniques discussed here are not popular for the growth of nanomaterials in comparison to chemical vapor deposition (CVD). The major reason is the tendency of forming thin films and amorphous by-products on the surfaces of catalysts which cause “poisoning” effect and prohibit the formation of nanotubes or nanowires [14]. However, there have been some pioneering works reported on the growth of nanotubes and

Physical Vapor Deposition, Fig. 3

(a) Low and (b) high magnification of gold nanoparticles deposited on one side of a boron nitride nanotubes (Adapted with permission from Lee, C.H. et al., *Adv. Mater.* **25**, 4544–4548 (2013). Copyright (2013) WILEY-VCH Verlag GmbH & Co)



nanowires by PVD techniques such as sputtering, PLD, and MBE. For example, vertically aligned carbon nanotubes (CNTs) were first successfully deposited by RF plasma-assisted pulsed laser deposition [15]. In this approach, two RF plasmas were employed. The first one was applied on the substrates to form a negative bias voltage during deposition. The second one was applied on a ring electrode located between the graphite target and the substrates. Hydrogen gas was used for the plasma generation, and Fe powders (20 nm in diameter) were employed as the catalysts. Although the density of these CNTs is low, these nanotubes have high structural order with internal nanotube diameter as small as 0.4 nm. The formation of these smallest possible CNTs was confirmed by transmission electron microscopy (TEM). Later, CNTs were deposited by magnetron sputtering with a dc bias voltage on substrates [16]. Prior to the growth of CNTs, Ni films (10 nm thick) were deposited by e-beam evaporation to form nanoparticles that later function as the catalyst for the vapor–liquid–solid (VLS) process. However, these CNTs are defective with bamboo-like structures. Later, RF plasma-assisted pulsed laser deposition was used to achieve low-temperature growth of boron nitride nanotubes (BNNTs) [14, 17]. In this case, pyrolytic boron nitride pallets were used as the targets, and a RF plasma was applied

on the substrates to create bias voltages, which are important to eliminate the formation of boron nitride thin films. Again, these BNNTs have internal nanotubes of 1 nm in diameter with high structural order.

Various PVD approaches were also employed for the synthesis of nanowires/nanorods. For example, ZnO and $\text{Zn}_{1-x}\text{Mg}_x\text{O}$ nanorods were grown by MBE on oxidized Si substrate coated with Ag catalyst [18]. In both cases, an ozone/oxygen mixture was used as the oxidizing source. The Zn and Mg cation fluxes were provided by a Knudsen effusion cell using high purity Zn and Mg metals as the sources. The growth of vertically aligned ZnO [19] and MgO [20] nanowires can also be obtained by PLD using ZnO or MgO targets, respectively. Gold particles were used as the catalysts for both cases.

More recently, PLD was shown to produce gold nanoparticles on the wide band gap (~ 6 eV) boron nitride nanotubes (BNNTs) [21]. These nanoparticles have functionalized the electrically insulating BNNTs to become the tunneling channel of tunneling field effect transistors (TFETs) without involving any semiconducting properties. Such transistors without the use of semiconductors are interesting as these TFETs by-pass many fundamental issues of semiconducting FETs such as leakage current, short channel effect, and high contact resistance

when the devices are scaled down to the quantum limits. Because of the related tunneling effect, these gold nanoparticles' functionalized BNNTs are called quantum dots' (QDs) functionalized BNNTs (QDs-BNNTs). As shown in Fig. 3, gold QDs smaller than 10 nm are preferentially coated on one side of the BNNT.

Acknowledgments Yoke Khin Yap acknowledges the support from the National Science Foundation (Award number DMR-1261910).

Cross-References

- Atomic Layer Deposition
- Carbon Nanotube-Metal Contact
- Carbon Nanotubes for Chip Interconnections
- Carbon-Nanotubes
- Chemical Vapor Deposition (CVD)
- Electron-Beam-Induced Deposition
- Focused-Ion-Beam Chemical-Vapor-Deposition (FIB-CVD)
- Graphene
- Growth of Silica Nanowires
- Nanomaterials for Electrical Energy Storage Devices
- Nanomaterials for Excitonic Solar Cells
- Nanotechnology
- Quantum Dots
- Synthesis of Carbon Nanotubes

References

1. Mattox, D.M.: *Handbook of Physical Vapor Deposition (PVD) Processing: Film Formation, Adhesion, Surface Preparation and Contamination Control (Hardcover)*. Noyes, Westwood, N.J (1998)
2. Mahan, J.E.: *Physical Vapor Deposition of Thin Films*. Wiley-Interscience, New York (2000)
3. Park, J., et al.: Epitaxial graphene growth by carbon molecular beam epitaxial (CMBE). *Adv. Mater.* **22**, 4140–4145 (2010)
4. Hackley, J., et al.: Graphitic carbon growth on Si (111) using solid source molecular beam epitaxy. *Appl. Phys. Lett.* **95**, 133114 (2009)
5. Tsang, W.T., et al.: Chemical beam epitaxy of InP and GaAs. *Appl. Phys. Lett.* **45**, 1234–1236 (1984)
6. Schiller, S., Jäsch, G.: Deposition by electron beam evaporation with rates of up to $50 \mu\text{m s}^{-1}$. *Thin Solid Films* **54**, 9–21 (1978)
7. Chrisey, D.B., Hubler, G.H. (eds.): *Pulsed Laser Deposition of Thin Films*. Wiley-Interscience, New York (1994)
8. Wang, J., Yap, Y.K.: Growth of adhesive cubic phase boron nitride films without argon ion bombardment. *Diamond Relat. Mater.* **15**, 444–447 (2006)
9. Yap, Y.K., Kida, S., Aoyama, T., Mori, Y., Sasaki, T.: Influence of negative dc bias voltage on structural transformation of carbon nitride at 600 °C. *Appl. Phys. Lett.* **73**, 915 (1998)
10. Yap, Y.K., Aoyama, T., Kida, S., Mori, Y., Sasaki, T.: Synthesis of adhesive c-BN films in pure nitrogen radio-frequency plasma. *Diamond Relat. Mater.* **8**, 382–385 (1999)
11. Yamamoto, K., Koga, Y., Fujiwara, S., Kokai, F., Heimann, B.: Dependence of the sp^3 bond fraction on the laser wavelength in thin carbon films prepared by pulsed laser deposition. *Appl. Phys. A* **66**, 115–117 (1998)
12. Iijima, S.: Helical microtubules of graphitic carbon. *Nature* **354**, 56–58 (1991)
13. Iijima, S., Ichihashi, T.: Single-shell carbon nanotubes of 1-nm diameter. *Nature* **363**, 603–605 (1993)
14. Wang, J., et al.: Low temperature growth of boron nitride nanotubes on substrates. *Nano Lett.* **5**, 2528–2532 (2005)
15. Yap, Y.K., Yoshimura, M., Mori, Y., Sasaki, T., Hanada, T.: Formation of aligned-carbon nanotubes by RF-plasma-assisted pulsed-laser deposition. Special issue for Tsukuba symposium on carbon nanotubes in commemoration of the 10th anniversary of its discovery; S. Iijima et al. eds. *Phys. B.* **323**, 341–343 (2002)
16. Lee, K.-F., et al.: Synthesis of aligned bamboo-like carbon nanotubes using radio frequency magnetron sputtering. *J. Vac. Sci. Technol. B* **21**, 1437–1441 (2003)
17. Xie, M., Wang, J., Yap, Y.K.: Mechanism for low temperature growth of boron nitride nanotubes. *J. Phys. Chem. C* **114**, 16236–16241 (2010)
18. Heo, Y.W., Kaufman, M., Pruessner, K., Norton, D. P., Ren, F., Chisholm, M.F., Fleming, P.H.: Optical properties of $\text{Zn}_{1-x}\text{Mg}_x\text{O}$ nanorods using catalysis-driven molecular beam epitaxy. *Solid-State Electron.* **47**, 2269–2273 (2003); Heo, Y.W., Varadarajan, V., Kaufman, M., Kim, K., Norton, D.P., Ren, F., Fleming, P.H.: Site-specific growth of ZnO nanorods using catalysis-driven molecular beam epitaxy. *Appl. Phys. Lett.* **81**, 3046–3048 (2002)

19. Rahm, A., et al.: Pulsed-laser deposition and characterization of ZnO nanowires with regular lateral arrangement. *Appl. Phys. A* **88**, 31–34 (2007)
20. Nagashima, K., Yanagida, T., Tanaka, H., Kawai, T.: Epitaxial growth of MgO nanowires by pulsed laser deposition. *J. Appl. Phys.* **101**, 124304 (2007)
21. Lee, C.H., et al.: Room-temperature tunneling behavior of boron nitride nanotubes functionalized with gold quantum dots. *Adv. Mater.* **25**, 4544–4548 (2013)

Introduction to Chemical Vapour Deposition

Being an effective way of constructing a wide range of components and products with potentially different material composition, chemical vapour deposition (CVD) has been developed since its inception as a novel manufacturing process in several industrial sectors. Notably, these include the semiconductor industry, ceramic industry and so forth. A trend of expanding its applications from initial mass semiconductor and microelectronics production to a wider range of applications has gained momentum in recent years as a result of intensive research and development work being undertaken by academic researchers and industrial end-users.

1.1 Definition of CVD

There are several definitions of CVD in the published literature. A practical and common definition of CVD is that it is a complex process of depositing solid materials at a high temperature as a result of a chemical reaction. This deposition forms a special type of material commonly known as ordered crystal grown from vapour.

Whilst the above definition introduces the basic high level understanding and observations of the process, a more concise and scientific definition for CVD is a process whereby a thin solid film is deposited onto a substrate through chemical reactions of the gaseous species. For structural component applications, the deposition typically takes place at a temperature of around 1000°C. It is the reactive processes that distinguish CVD process from physical vapour deposition (PVD) processes, such as physical evaporation process, sputtering and sublimation processes [1].

Figure 1.1a gives a typical example of a CVD system, where reactant gases, normally called precursor gases of CH_3SiCl_3 and H_2 , are delivered into a reaction chamber at a suitably determined temperature. As they pass through the reactor these gases come into contact with a heated substrate; they then react and form a solid SiC layer deposited onto the surface of a substrate. Usually, an inert gas, such as Ar, is used as a diluent gas. The depositing temperature and pressure are the critical parameters in this process. After the reactions, the exhaust gases containing HCl species are trapped by NaOH and then condensed by liquid N_2 trap before being released into the atmosphere.

CVD is therefore a generic name for a group of processes that involve forming a thin layer via chemical reaction and depositing a solid material layer onto substrates. Figure 1.1b shows a model consisting of the sequential physical and chemical steps that occur during a CVD process, developed by Spear in 1982 [2]. They are summarised as follows:

1. Mass transport of reactant gaseous species to vicinity of substrate;

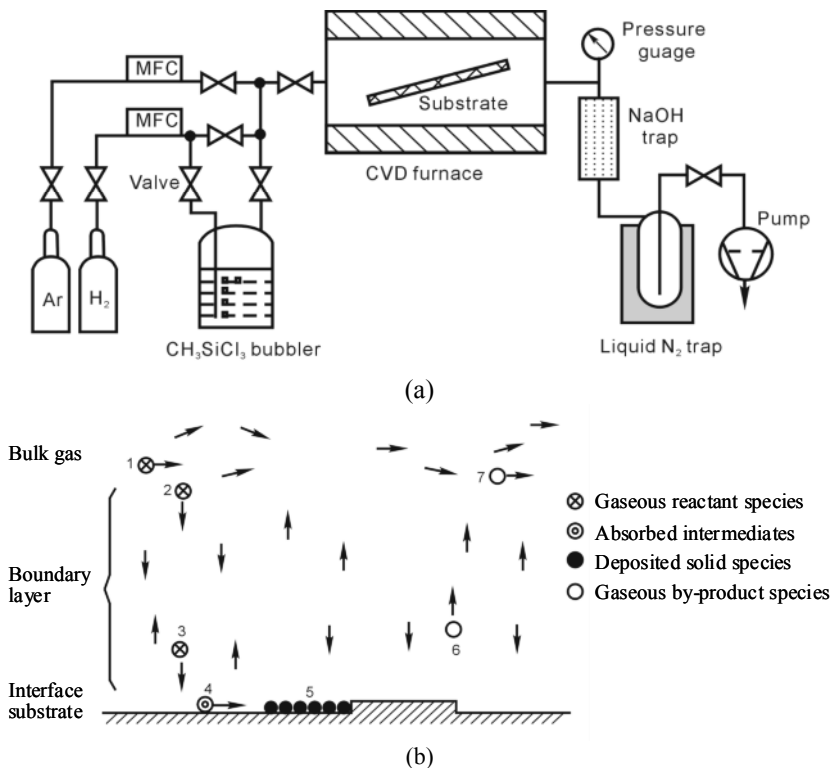


Figure 1.1. Schematics of a CVD process: (a) arrangement of a CVD system and (b) a CVD model

2. Diffusion of reactant species through the boundary layer to the substrate surface or homogeneous chemical reactions to form intermediates;
3. Adsorption of reactant species or intermediates on substrate surface;
4. Surface migration, heterogeneous reaction, inclusion of coating atoms into the growing surface, and formation of by-product species;
5. Desorption of by-product species on the surface reaction;
6. Diffusion of by-product species to the bulk gas; and
7. Transport of by-product gaseous species away from substrate (exhaust).

1.2 Characteristics of Chemical Vapour Deposition

Mechanical manufacturing techniques have been developed based on different working principles and component manufacturing techniques. These can be classified into two main categories, namely additive methods and subtractive methods (see Figure 1.2), using the method of component formation as a criterion. Traditional manufacturing techniques, such as milling, turning, drilling, grinding and cutting, rely on processes in which a component is gradually formed by

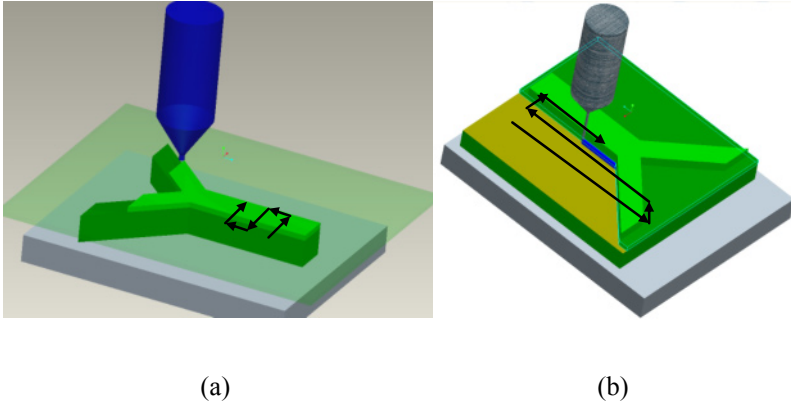


Figure 1.2. Components made by (a) additive method and (b) subtractive method

removing or subtracting materials from a simple shaped and larger block of uniform material by casting, forging, rolling or extruding, usually called a work piece. These techniques are hence called subtractive methods as materials are gradually subtracted or removed from the work piece.

Recent new techniques have been developed to gradually build up a component by adding more material in solid powder form followed by sintering or soldering, in fluid form followed by solidification, or in gaseous form followed by chemical reactions or physical deposition. These techniques mostly work on the so-called layer manufacturing principle. A close control of the layer-based addition process is therefore required to achieve the final dimensions, accuracy and surface finishes.

CVD processes work on the principle of adding a new layer of material onto a substrate surface and hence belong to the additive manufacturing technique family. They employ a suitable substrate as the base component and gradually deposit new material onto the base to form a component. In contrast to other additive manufacturing techniques, CVD processes do not require any external driver or causes for their solidification processes, such as laser-based solidification found in stereolithography technique, or spread binders in powder-based 3-D printing. They rely on chemical reactions to form a strong bond for deposits. These reactions occur at the atomic level; hence the deposition formed on a substrate can be very fine and be comprised of thin layers of coating. CVD processes can therefore be considered an accurate manufacturing technique at the micro or even nano level.

CVD has a number of advantages as a method for depositing layers of thin-film materials onto a substrate.

1. Quite different from some physical vapour deposition methods (PVD), CVD is a non-line-of-sight process which leads to the good conformality in term of uniform thickness of the coating. Figure 1.3 (a) shows an example of conformal coverage of a step shaped substrate coated by a CVD process. CVD also has high throwing power which is exceeded only by plasma spraying. As a result, the coatings can be uniformly deposited over substrate contours and complex surfaces. This means that the coatings can

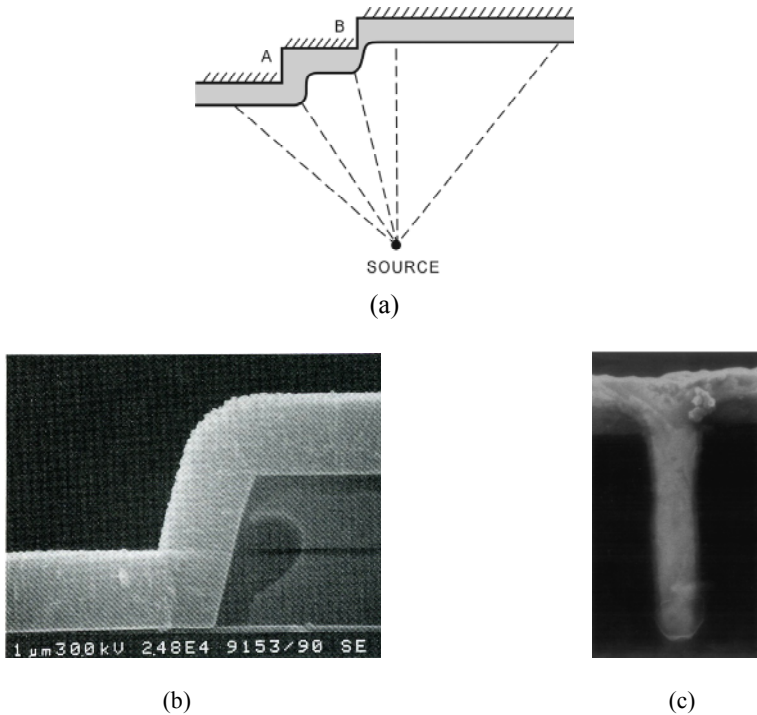
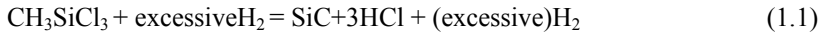


Figure 1.3. Comparison between PVD and CVD processes: (a) PVD coating on a step substrate, (b) CVD TiN on a step Si substrate [3] and (c) CVD Cu into a micro-trench [4]

be applied to elaborately shaped work pieces, including those with the interior and underside features, and those with high aspect ratio holes and other similar features which can be completely filled. Deep recesses, holes, and other difficult three-dimensional features can usually be coated with relative ease by the CVD process. For instance, a micro-trench with an aspect ratio of 10:1 can be completely filled with copper using CVD technique, as shown in Figure 1.3 (b). In contract, PVD techniques, such as sputtering or evaporation, generally require a line-of-sight between the surface to be coated and the source, e.g. the coating source should have a straight line path to the surface on the substrate, as shown in Figure 1.3 (c).

2. CVD processes have a greater flexibility of using a wide range of chemical precursors such as halides, hydrides, organo-metallic compounds and so forth which enable the deposition of a large spectrum of materials, including metals, non-metallic elements, carbides, nitrides, oxides, sulphides, as well as polymers. Up to now, around 70% of elements in the periodic table have been deposited by the CVD technique, some of which are in the form of the pure element, however, more often the compound materials.

3. The CVD technique requires relatively low deposition temperatures. This effectively enables the desired materials to be deposited in-situ at low energies through vapour phase reactions. This enables the deposition of refractory materials at a temperature much lower than their melting temperatures. For example, refractory materials such as SiC (sublimation point: 2700 °C) can be deposited at about 1000 °C using the following chemical reaction:



Furthermore, the deposition temperature could be significantly decreased to around 300 °C with the enhancement techniques such as by using plasma, photo, or laser tools.

4. CVD also has the ability to control the crystal structure, stoichiometry, surface morphology and orientation of the CVD manufactured products by tailor-making through controlling the deposition parameters. Figure 1.4 shows two structures produced using the CVD technique, which illustrate a high quality formation of multilayered coatings composed of HfC/SiC for the image on the left and PyC/SiC on the right.
5. The deposition rate can be adjusted readily. Low deposition rate is favoured for the growth of epitaxial thin film for microelectronic applications. However, for the deposition of thick protective coatings, a high deposition rate is preferred and it can be greater than tens of micrometers per hour. The deposition rate using CVD technique is high and thick coatings can be readily obtained (in some cases centimetres in thickness). The process is generally competitive compared with other techniques and, in some cases, CVD is more economical than PVD processes.
6. Compared with the facilities used in PVD, the CVD equipment does not normally require ultra-high vacuum working environments and the equipment generally can be adapted to many process variations. This great flexibility is advantageous such that it allows many changes in composition

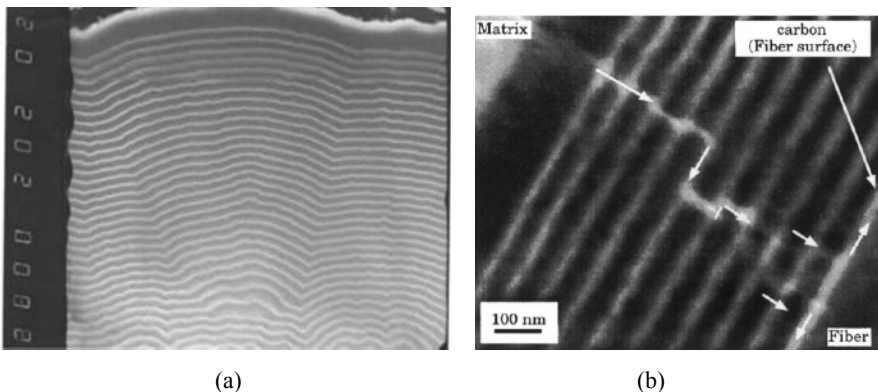


Figure 1.4. Multilayered coatings by CVD processes: (a) HfC/SiC [5] and (b) PyC/SiC [6]

during the deposition, and the co-deposition of elements or compounds into multiple layers or mixed layers can be readily achieved using the CVD technique.

However, CVD processes also have several disadvantages as follows:

1. A major disadvantage is that CVD requires chemical precursors as the essential reactants, which can often present safety and health hazards as they can be at times extremely toxic, corrosive, flammable and explosive. CVD precursors can be highly toxic even at low concentration (e.g. $\text{Ni}(\text{CO}_4)$), explosive such as B_2H_6 , or corrosive (e.g. SiCl_4). Some precursors, especially the metal-organic precursors, can be quite expensive to use. The exhaust gases consist of chemical reaction by-products, intermediates and others, such as CO , H_2 or HCl , which can also be hazardous, toxic, corrosive and flammable. These gases must be properly treated before they are released into the environment. The treatment processes include neutralisation, condensation and filtering etc, which may be very costly and environmentally controversial.
2. Compared with other vapour-phase deposition methods, CVD method is perhaps the most complex. Unlike growth by physical deposition such as evaporation or Molecular Beam Epitaxy (MBE), this method requires numerous test runs to determine and reach suitable growth parameters, especially for single-crystal growth. The complexity of this method results from the following facts:
 - the chemical reactions generally involve multiple gaseous species and produce a number of intermediates;
 - during the above complex reactions, it is extremely difficult to identify the reaction processes and detect the resultant intermediates in most cases;
 - for a CVD, there are a sequence of chemical reaction steps for deposition compared with much simpler physical methods, and
 - CVD process is also difficult to control and requires a lot of experience and tests before a reliable control algorithm can be fully developed to control the gas which has greater freedom to flow in a typical deposition chamber.

Compared with other material forming methods such as ceramic sintering and liquid metal casting, CVD shows much more challenges and difficulties in devising effective and flexible control methods due to the above reasons.

1.3 Overview of CVD Development

As an important step-change and disruptive technology, the CVD has been developed since 1960s into a relatively mature and reliable manufacturing technique. The development of CVD could be divided into four key eras, based on the maturity and the application nature of the technology:

1.3.1 Stage 1: Early Development Era

The formation of soot in a furnace due to incomplete oxidation of firewood during prehistoric times is probably the first observation, by human being, of deposition using a CVD process. The development of the above observation is in common with many technology developments and has been closely linked to the practical needs of societies. Later, this method was used to produce carbon black pigment from the soot and prepare pre-historic art on the walls.

A more recent application and the industrial exploitation of CVD could be traced back to a patent filed by de Lodyguine in 1893[7, 8]. His invention was to deposit Tungsten (W) onto carbon lamp filaments through the reduction of WCl_6 by H_2 . This advancement transferred the CVD from a trivial technique used to make colour pigment into a major industrialisation technique which benefited human beings significantly by producing reliable, more accessible and cheaper light bulbs.

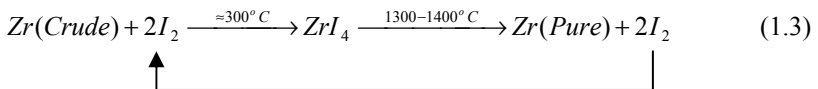
1.3.2 Stage 2: Refining or Purification of Metals Era

Following the initial deposition applications of metals, CVD method was further developed and widely used mainly for refining or purification of some metals (especially metals such as Ni, Zr, Ti etc) until the 1940s.

One of the earliest industrial CVD applications developed during this stage is a carbonyl process for refining nickel (Ni), developed by Mond in 1890 [9, 10]. The famous Mond extraction process uses the following chemical reaction process:



Later, a new extraction process was developed by Anton Eduard van Arkel and Jan Hendrik de Boer in 1925, known as van Arkel and De Boer process [11, 12], and the process uses the following chemical reaction:



The van Arkel-de Boer process has been used to extract Zr, Ta, etc. The extraction principle is explained through the schematic diagram of Zr extraction process and associated equipment, as shown in Figure 1.5. This process uses a U-shaped hot filament connected to the tungsten electrodes, and the temperature of the heating element surface can reach to around 1300 – 1400 °C. In zirconium (Zr) purification process, Zr heating filament is the best choice although W or Mo can be used. At this temperature, zirconium from zirconium crude metal attaches to the chamber interior wall whose temperature is around 300 °C, reacts with iodine (I_2) from its container to form ZrI_4 in gas state, which is then transported to the hot filament due to its concentration gradient. ZrI_4 is then decomposed into Zr and I_2 at 1300 – 1400 °C, resulting in zirconium depositing onto the surface of the hot filament. Using this way, pure zirconium is produced, whereas iodine returns back to the crude Zr

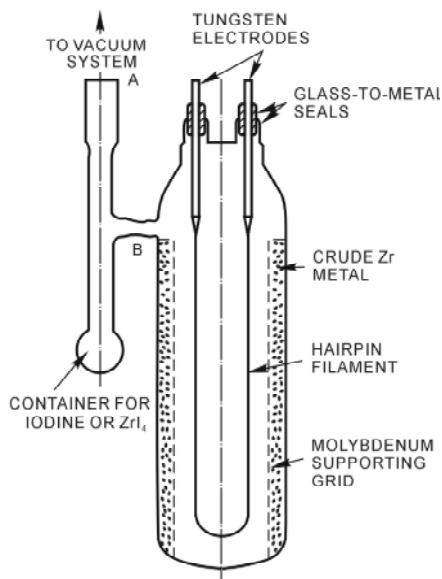
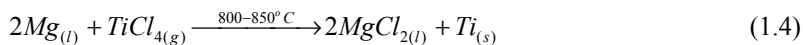


Figure 1.5. Schematic for van Arkel and De Boer process [13]

area to react in the next cycle with the crude zirconium metal. This cyclic and repetitive reaction between iodine and crude zirconium metal continues to produce more ZrI_4 , which in turn is decomposed at the hot filament and pure zirconium is deposited further onto the hot filament surface. The above process continues till all crude zirconium metal is converted into pure zirconium. During the above process, many other impurities including oxides and nitrides do not react with iodine, whereas metal impurities (e.g. Fe, Cr, Ni and Cu etc) react with iodine, but their resultant products are vaporised very slowly under the above process conditions. The above processes help to produce high-grade pure zirconium.

Another typical process called Kroll process was developed by Kroll in 1940s [14, 15]. This process is still being widely employed to manufacture titanium (Ti) by magnesium reduction of the titanium tetrachloride through the following reaction:



It was also during this period that silicon was first deposited using a method of hydrogen reduction of silicon tetrachloride by Pring and Fielding in 1909 [16]. The development and availability of this particular technique made it possible to develop and manufacture microelectronic devices in a large scale.

1.3.3 Stage 3: Microelectronics Manufacture Era

The evolution of the CVD technique had not reached a critical and significant stage until the rapid development of demands and corresponding technologies became a main industrial driving force to start electronic revolution in 1960s [17, 18]. It is only in the past 40 years that a considerable in-depth understanding of the process has been made and the increased applications of CVD explored. During this period, CVD technology evolved as a key technology to produce micro scale electronic components, driven by the demands of miniaturisation of computer components. The ultra-fine scale deposition characteristics of a CVD process possesses a desirable capability to produce super fine films required for semiconductors and other applications. In early 1970s CVD became a successful technique to manufacture electronic semiconductors and protective coatings for electronic circuits. This significantly reduces the size of electronic components by replacing large electric vacuum tubes with silicon circuits which integrate many individual electric components onto a single silicon wafer. Figure 1.6 shows a typical schematic diagram of a MOS field effect transistor, in which the poly-silicon, dielectric SiO_2 and Si_3N_4 are produced by the CVD processes [19].

In manufacturing microelectronic devices, CVD processes are used to fabricate several kinds of films made of the following materials [18]:

1. Active semiconductor materials, including
 - group IV doped elements and compounds such as Si, Ge, SiGe and SiC;
 - group III-V compounds such as AlN, AlAs, AlP; GaN, GaP, GaAs; InP, InAs; AlGeAs, GaAsP, GaAsSb, GaInP, GaInAs, InAsP, as well as GaInAsP;
 - group II-VI compounds such as ZnS, ZnSe, ZnTe, CdS, CdSe and CdTe;
 - miscellaneous semiconductors: ScN, YN, SnO_2 , In_2O_3 , PbSnTe, etc.
2. Conductive interconnect materials:
 - some element metals and alloys: Ag, Al, Al_3Ta , Au, Be, Cu, Ir, Mo, Nb, Pt, Re, Rh, V, Ta, Ta-W, W, W-Mo-Re, etc.
3. Insulating dielectric materials:
 - aluminosilicate glass (AISG) deposited from $\text{SiH}_4\text{-Al}(\text{CH}_3)_3\text{-O}_2\text{-N}_2$; $\text{Si}_3\text{N}_4\text{-Al}$ etc. $\text{Si}_x\text{O}_y\text{N}_z$, etc.
 - arsenosilicate glass (AsSG) deposited from $\text{Si}(\text{OC}_2\text{H}_5)_4\text{-AsCl}_3\text{-O}_2\text{-N}_2$;
 - borosilicate glass (BSG) deposited from $\text{SiH}_4\text{-B}_2\text{H}_6$ system;
 - phosphorsilicate glass (PSG) deposited from $\text{SiCl}_4\text{-PCl}_3\text{-H}_2\text{O}$, $\text{PH}_3\text{-SiH}_4\text{-O}_2$;
 - oxides SiO_2 , Al_2O_3 , TiO_2 , ZrO_2 , HfO_2 , Ta_2O_5 and Nb_2O_5 ;

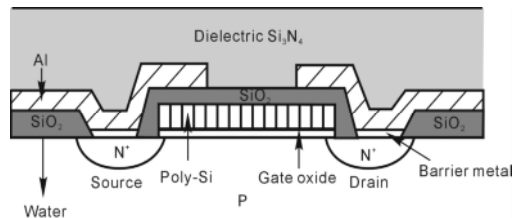


Figure 1.6. Schematic of an n-type metal-oxide-semiconductor (MOS) field transistor cross-section [19]

The research and development of CVD in recent years have focused on solid state microelectronic devices. This rapidly moving technology demands continuously improved materials and processes for the fabrication of even more advanced semiconductor devices. It is well recognized that computer chips would not be manufactured in their current capacity and structure if CVD techniques were not transferred and developed from the material extraction method into the deposition technology.

1.3.4 Stage 4: Wider Applications Era

In recent years, a great deal of success has been achieved and CVD techniques have been widely applied and further developed because of the following enablers and reasons [18].

The market needs for more advanced materials provide the driving force of the CVD processes. For example, the development of high performance aero-engines requires the strong and tough fibre reinforced ceramic matrix composites with low density which can potentially be used in ultra-high temperatures.

With more advanced instruments available, it is possible to identify the reactant gaseous species of the chemical reactions. This in turn helps to improve the understanding of the underlying chemical reactions and the processes;

The sophisticated and more efficient CVD systems can be designed and constructed with the aid of computational fluid dynamics (CFD) and high performance computers. At the same time, control technology has advanced such that it becomes much easy to accurately control the CVD processing parameters, such as temperature, pressures and flow rate, etc;

Tremendous fundamental research especially in the CVD technologies and precursor synthesis extends the applicability to some new area. These technologies include metal-organic CVD (MOCVD), plasma-enhanced CVD, photo CVD and laser induced CVD etc. The advancement of the synthesis technology provides new precursors to deposit a variety of materials with high purity.

Based on the above advancement, CVD processes have been widely used to manufacture various semiconductor devices, MEMS, nanomaterials, and advanced structural materials for ultra-high temperature applications. Some of these applications are briefly introduced and summarized as follows.

1. Communication industry [20]
 - Complex epitaxial hetero-structures of SiGe or compound semiconductor (e.g. AlGaAs) for high frequency (1 – 100 GHz) applications;
 - High purity SiO₂ optical fibre for the desired refractive index profile;
2. Optoelectronic [21, 22]
 - High brightness blue and green LEDs based on group-III nitride alloys (e.g. InGaN).
 - The light emitting diodes (LEDs) have been widely used in recent years as a general source of white light. They exhibit many desirable features as a light source, namely, high luminescence efficiency, fast response time, and reliable long service life. Red LEDs have been made widely available, but the high intensity blue and green LEDs were unavailable before 1990s. As

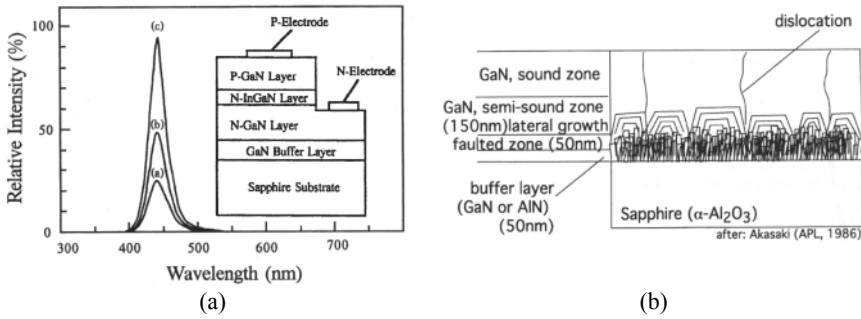


Figure 1.7. GaN blue LEDs by MOCVD: (a) Electroluminescence spectra and (b) Structure of a buffer layer [22]

blue is one of the three primary colours, the development such a LED is required to produce full colour spectrum and pure white light. In his well recognised attempt in 1993, Nakamura and his group used metal-organic CVD technique to develop GaN and related semiconductors which are the key components for the blue LEDs. Figure 1.7 shows the electroluminescence wavelength and its structural configuration of a blue LED. It is believed that GaN-based LEDs may well soon replace incandescent light bulbs and fluorescent tubes in a similar way as silicon circuits replaced vacuum tubes in the past.

3. Microelectromechanical structures (MEMs) [23, 25]

- Complex and smart micro-components with the size as small as 50 – 200 μm

4. Nanotechnology

- Some nano-materials and nano-devices fabricated primarily for functional applications, such as nanopowder [26], nanowire [27], nanobelt [28], nanotube [29] as well as nanocoil [30].

5. Coatings

- Diamond-like carbon (DLC) coatings [31, 32]
- SiC for oxidation protection for carbon-based composites [33]
- Refractory metal coatings [34], such as Re/Ir coating for the oxidation resistance and gas tightness of composite chamber at ultra-high temperatures
- hard coating (TiN, B_4C , TiC and Ti–Si–C–N) for the anti-wear of cutting tools [35], [36]
- Corrosion protection coating for gas turbines [37, 38]
- Reflection mirror, such as SiC [39]

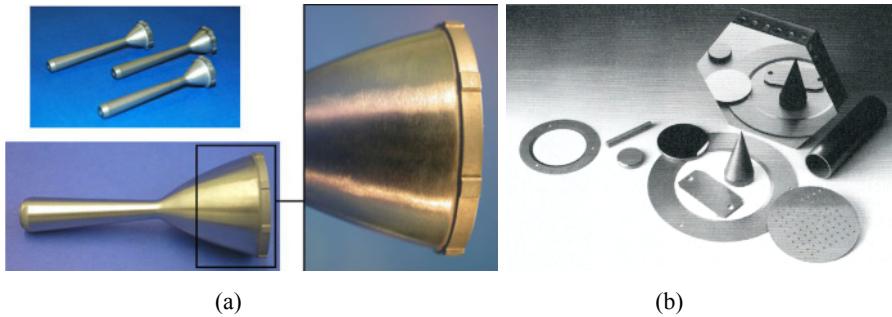
6. High temperature (T_c) superconductors [40]

$\text{YBa}_2\text{Cu}_3\text{O}_{7-x}$ and Bi–Sr–Ca–Cu–O oxides

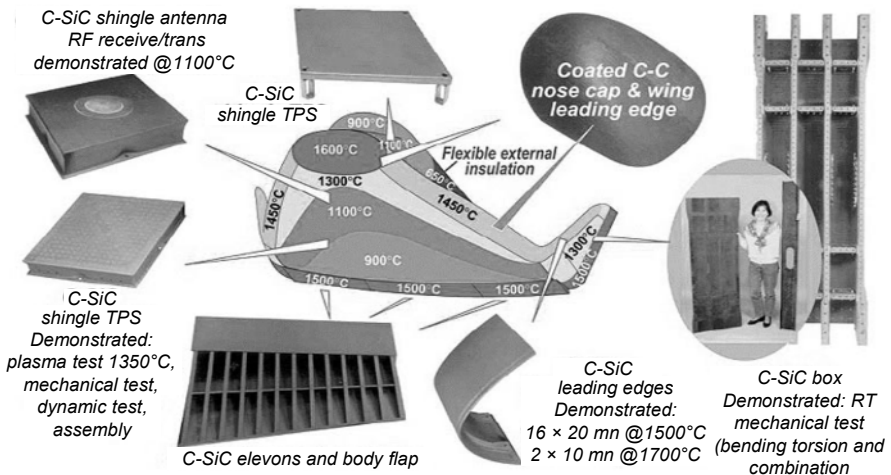
7. High-purity bulk materials [41, 42] ZnS, ZnSe, CdS, CdTe, SiC, Si_3N_4 , B_4C , GaAs, Rh and W etc. Some of the components are as shown in Figure 1.8.

8. Refractory composites

- high-performance ceramic fibres: SCS–SiC and B monofilaments [43, 44];



- carbon/carbon composite for friction and ultra-high-temperature applications in aircraft braking systems[45];
- carbon/silicon carbide composites for thermal protection systems of space shuttles [46], as shown in Figure 1.9;
- silicon carbon/silicon carbide composite components for long service-time in aero-engines with high thrust/weight ratio [47] as well as the first wall used in a nuclear fission reactor [48];
- carbon/(carbon-carbide) multi-matrix composites for ablation resistance [49].



1.4 High-performance Ceramic Matrix Composites and Chemical Vapour Infiltration

CVD has also been developed into an advanced version named Chemical Vapour Infiltration and has found an important application of making higher performance ceramic materials. Compared with super-alloys, advanced ceramics exhibit superior performances which include very low density, high strength and excellent erosion resistance to the harsh service environments at elevated temperatures. However, the brittleness causing low reliability is the key problem, limiting their wide range of applications. Research on how to obtain tough advanced ceramics has received considerable interests. As illustrated in Figure 1.10, the development of tough ceramics can be divided into three stages [50]:

In the middle of 1960s, the fracture mechanics was developed to a stage where it was possible to apply it to the new applications. The development of the indentation fracture helps to focus the design attention on flaw elimination in ceramics, which aim to achieve the highest degree of homogeneity and density in monolithic ceramics. As a result, dense ceramics were produced and they possess higher mechanical strength, but they are still brittle.

The great success of tough ZrO_2 ceramics inspired material scientists to develop new materials by design. This should establish a closer relationship among processing, characterizing, testing, modelling and so forth. Especially, the mechanical properties of ceramics can be significantly improved by designing its microstructure through the incorporation of whisker, elongated crystal, platelet, and fibre.

Fibre-reinforced ceramic matrix composites with high workload for fractures were firstly demonstrated in the early of 1970s. Vigorous activities were pursued after high performance Nicalon/glass-ceramic matrix composites were reported by Prewo and Brennan [51, 52], and carbon/silica glass as well as carbon/silicon nitride by Guo [53, 54] in the early 1980s.

Various methods have been developed to fabricate the tough ceramics, which

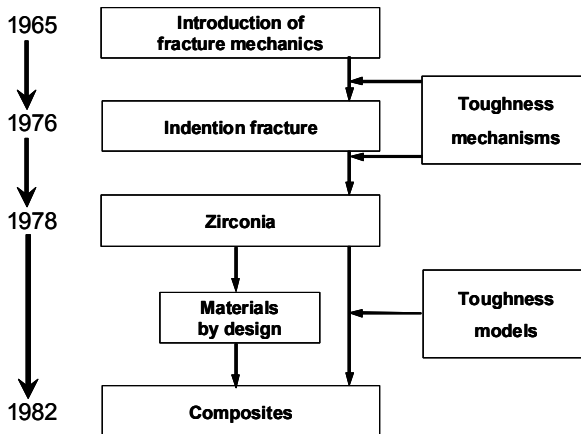


Figure 1.10. Overall chronological developments of high-toughness ceramics [50]

Table 1.1. Comparison among various toughening methods for ceramics [50]

Mechanism	Highest toughness (MPa·m ^{1/2})	Materials
Transformation	~20	ZrO ₂ /Al ₂ O ₃ , Ce-TZP, Y-TZP
Microcracking	~10	Al ₂ O ₃ /ZrO ₂ , Si ₃ N ₄ /SiC, SiC/TiB ₂
Metal dispersion	~25	Al ₂ O ₃ /Al, ZrB ₂ /Zr, Al ₂ O ₃ /Ni, WC/Co
Whiskers/platelets	~15	Si ₃ N ₄ /SiC, Si ₃ N ₄ /Si ₃ N ₄ , Al ₂ O ₃ /SiC
Self toughing	12	In-situ Si ₃ N ₄ [55], LaB ₆ -ZrB ₂ [56], Al ₂ O ₃ /Y ₃ Al ₅ O ₁₂ , Al ₂ O ₃ /GdAlO ₃ [57]
Fibres	≥30	SiC/LAS, SiC/CAS, SiC/MAS, C/BAS, SiC/SiC, C/SiC [58]

*LAS: Li₂O–Al₂O₃–SiO₂, CAS: CaO–Al₂O₃–SiO₂
MAS: MgO–Al₂O₃–SiO₂, BAS: Li₂O–Al₂O₃–SiO₂
Y-TZP: tetragonal zirconia polycrystal stabilised by Y₂O₃

are listed in Table 1.1. Among them, fibre reinforcement is considered as the most promising method. In 1986, the outstanding mechanical properties of the CVI SiC/SiC composites were systematically reported by Lamicq *et al.* As shown in Figure 1.11, the CVI SiC/SiC composites exhibited excellent fracture behaviours. The fracture toughness can be as high as 25 MPa·m^{1/2} from room temperature up to 1400 °C and a very high level of fracture work of more than 10 kJ·m⁻² is required.

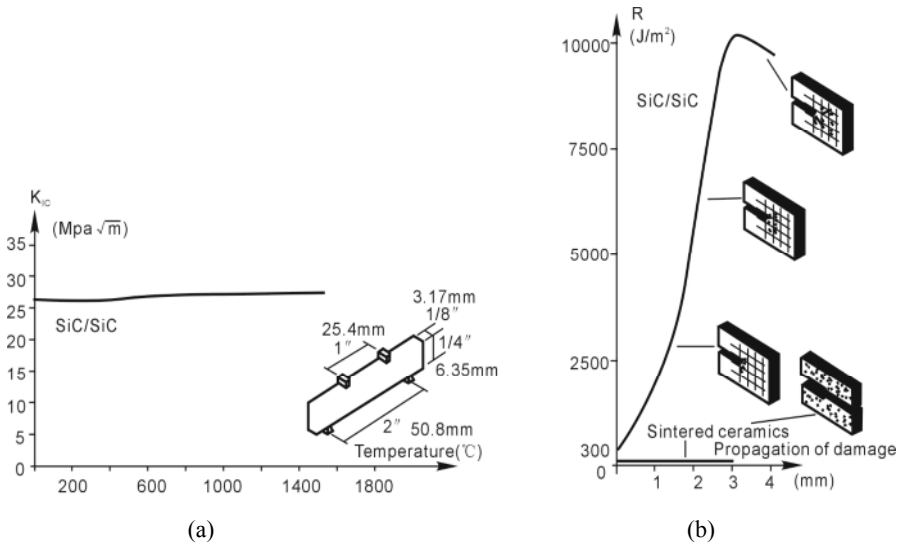


Figure 1.11. Mechanical properties of 2-D SiC/SiC composites [59]: (a) fracture toughness and (b) fracture resistance curve

In this method, a thin layer between fibre and matrix with proper interfacial bonding plays critical role for the crack deflection and fibre pull-out as shown in Figure 1.12. The CVI technique possesses outstanding advantages to deposit thin coatings on the surface of individual fibres within the preform then followed by the subsequent infiltration of the matrix phase.

Chemical vapour infiltration (CVI) is an extension of CVD processes only when a CVD process occurs on an internal surface of a porous substrate (especially for the fibre preform). As compared with CVD, the CVI process for ceramics is much more effective and important because it is the optimal technique to fabricate fibre reinforced ceramics and particularly carbon fibre reinforced carbon and advanced ceramic matrix composites. Both CVI and CVD techniques share some common features in overall chemistry, however, the CVI is much more complex than the CVD process in mass transport and chemical reactions.

For a CVD process, the mass transport and chemical reactions occur only on the surface of a dense substrate, producing a thin layer of coating. For a CVI process, the reactant gases must be transported into the porous fibre preform, in which the size of the pores varies with a large distribution, e.g. from the large pores ($\sim 10^2 \mu\text{m}$) among the fibre bundles to the small pores ($10^0 \sim 10^{-1} \mu\text{m}$) among the fibres within a bundle. The homogeneous chemical reactions are therefore competing with the heterogeneous chemical reactions within the pores, resulting in a complex deposition process. The mass of the deposit in a component made from a CVI process is two to ten times greater than that of the fibre preform because the deposit becomes the main constituent of the composites. In order to obtain the required density and improve the density uniformity of the composites, some special CVI techniques have been developed, including isothermal/isobaric CVI, forced CVI, thermal gradient CVI, liquid immersion CVI, pulsed CVI, and so forth. Up to now, CVI techniques have become the commercial method to manufacture the fibre reinforced ceramic matrix composites and carbon/carbon composites.

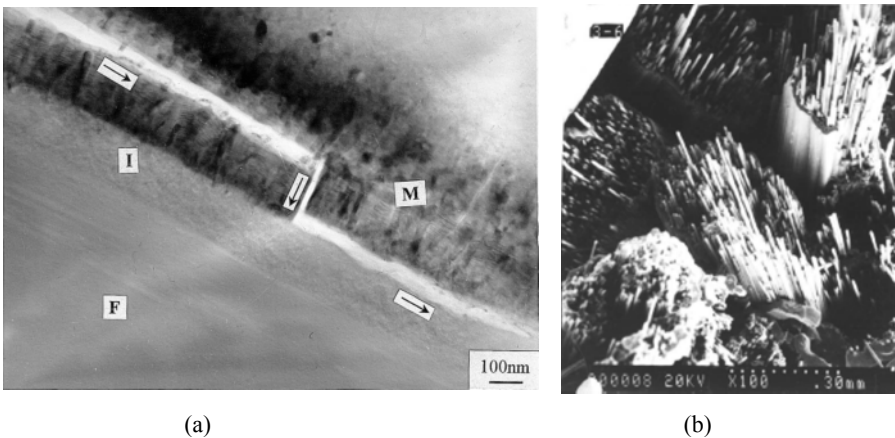


Figure 1.12 Crack deflection and fibre pull-out in 3-D C/SiC composites: (a) crack deflection along the interface [60] and (b) fibre pull-out [58]

1.5 Literature Sources

Whilst this chapter and indeed the book aim to provide a clear and insightful introduction to the complex CVD subject area, it is necessary to cover both theoretical and application based topics. This is because that CVD itself has evolved for many years into different applications and it is also built on evolving related theoretical developments. As the emphasis of the book is to provide an in-depth explanation of the subject from engineering application points of view, the book hence will have a light touch on the development of related theories as well as non-main stream applications. In order to compensate this slight lack of introduction to theories due to the above approach, a list of comprehensive literature references has been compiled to further the understanding of the theory, particular applications and more specialised topics. The literatures on the subject of CVD have been structured into the categories of books, handbooks, journals, conference proceedings and general representative papers.

From these references, it is not difficult to see that chemical vapour deposition has been advanced based on a very extensive literature of books, journals and conference proceedings with various aspects of the subject. The list in this section aims to provide a comprehensive source of readings of the most recent and relevant references for readers to pursue the subject further.

1.5.1 Books

- *Vapour Deposition*, Powell, Oxley and Blocher Jr, 1966 [61]
- *Thin Film Processes*, Vossen and Kern, 1978 [62]
- *Chemical Vapour Deposition*, Hawkins, 1981 [63]
- *Chemical Vapour Deposition for Microelectronics*, Sherman, 1987 [64]
- *Thin Films by Chemical Vapour Deposition*, Morosanu, 1990 [65]
- *Thin Film Processes II*, Vossen and Kern, 1991 [66]
- *The Materials Science of Thin Films*, Ohring, 1992 [67]
- *Chemical Vapour Deposition: Principles and Applications*, Hitchman and Jensenin, 1993 [1]
- *The Chemistry of Metal CVD*, Kudas and Hampden-Smith, 1994 [68]
- *CVD of Nonmetals*, Rees, 1996 [69]
- *CVD of Compound Semiconductors*, Jones and O'Brien, 1997 [70]
- *Chemical Vapour Deposition: Thermal and Plasma Deposition Electronic Materials*, Sivaram, 1995 [71]
- *Chemical Vapour Deposition*, Park and Sudarshan, 2001 [72]

1.5.2 Handbooks

- *Handbook of Thin Film Deposition Processes and Techniques*, Schuergraf, 1988 [73]



A brief review of atomic layer deposition: from fundamentals to applications

Richard W. Johnson^{1,3}, Adam Hultqvist^{2,3} and Stacey F. Bent^{1,2,*}

¹ Department of Materials Science and Engineering, Stanford University, Stanford, CA 94305, USA

² Department of Chemical Engineering, Stanford University, Stanford, CA 94305, USA

Atomic layer deposition (ALD) is a vapor phase technique capable of producing thin films of a variety of materials. Based on sequential, self-limiting reactions, ALD offers exceptional conformality on high-aspect ratio structures, thickness control at the Angstrom level, and tunable film composition. With these advantages, ALD has emerged as a powerful tool for many industrial and research applications. In this review, we provide a brief introduction to ALD and highlight select applications, including Cu(In,Ga)Se₂ solar cell devices, high-k transistors, and solid oxide fuel cells. These examples are chosen to illustrate the variety of technologies that are impacted by ALD, the range of materials that ALD can deposit – from metal oxides such as Zn_{1-x}Sn_xO_y, ZrO₂, Y₂O₃, to noble metals such as Pt – and the way in which the unique features of ALD can enable new levels of performance and deeper fundamental understanding to be achieved.

Introduction

Atomic layer deposition (ALD) is a technique capable of depositing a variety of thin film materials from the vapor phase. ALD has shown great promise in emerging semiconductor and energy conversion technologies. This review is intended to introduce the reader to the basics of ALD and highlight current applications pertaining to microelectronics and energy that were selected because of their importance in either industry or research. For a more comprehensive summary of ALD and its many applications, the reader is referred to existing reviews on the topic [1–10].

As device requirements push toward smaller and more spatially demanding structures, ALD has demonstrated potential advantages over alternative deposition methods, such as chemical vapor deposition (CVD) and various physical vapor deposition (PVD) techniques, due to its conformality and control over materials thickness and composition. These desirable characteristics originate from the cyclic, self-saturating nature of ALD processes.

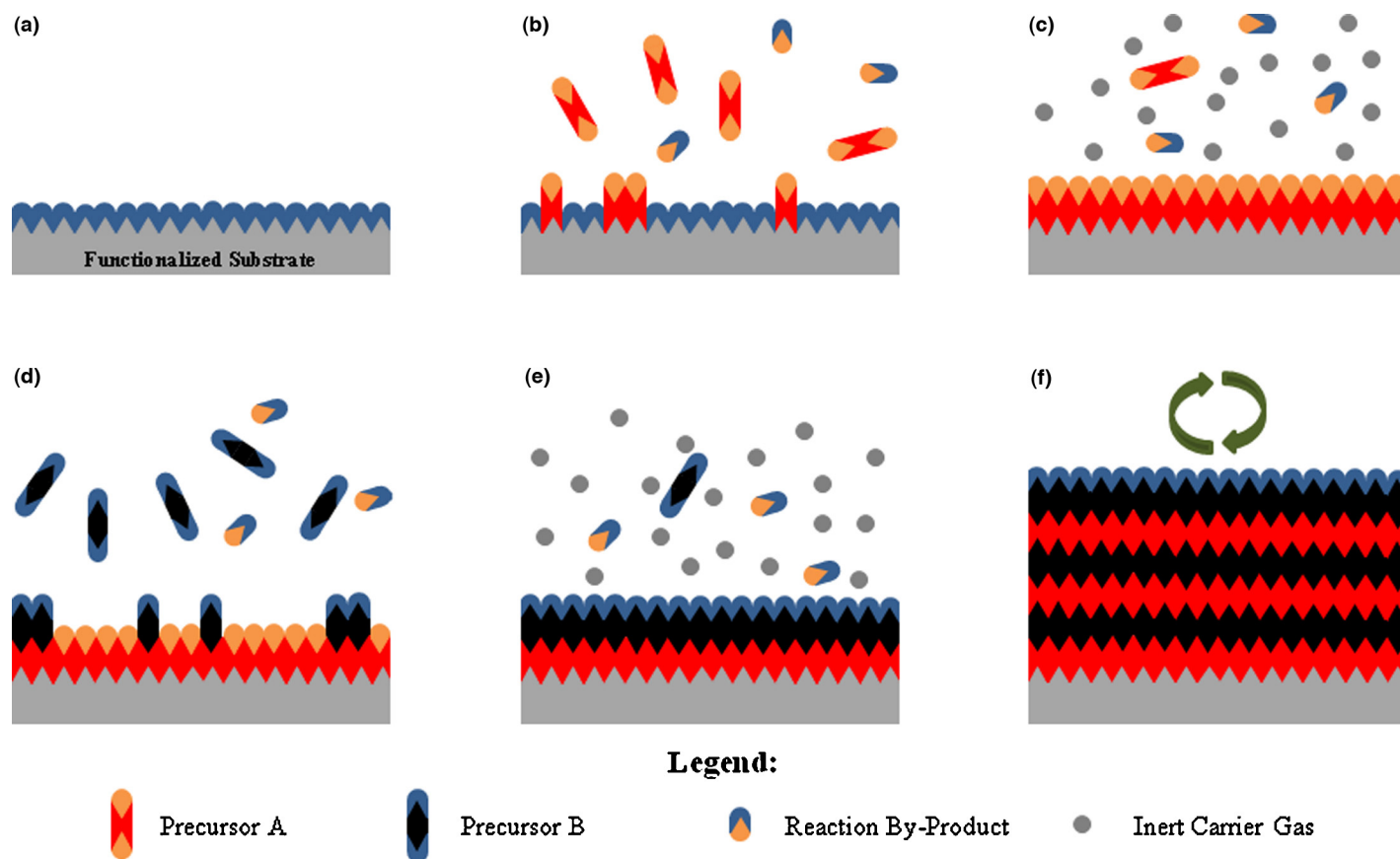
ALD was popularly introduced as atomic layer epitaxy (ALE) by Suntola and Antson in 1977, depositing ZnS for flat panel displays

[11]. As further ALE processes were developed to incorporate metals and metal oxides, many materials were deposited non-epitaxially and the more general name of ALD was adopted to reflect this [1]. It should be noted, too, that many ALD procedures were developed from a variety of CVD processes. In contrast to their CVD analogs, the ALD procedures feature alternating exposure of chemical precursors to react to form the desired material, often at significantly lower temperatures [12].

A general ALD process is illustrated in Fig. 1. It consists of sequential alternating pulses of gaseous chemical precursors that react with the substrate. These individual gas-surface reactions are called ‘half-reactions’ and appropriately make up only part of the materials synthesis. During each half-reaction, the precursor is pulsed into a chamber under vacuum (<1 Torr) for a designated amount of time to allow the precursor to fully react with the substrate surface through a self-limiting process that leaves no more than one monolayer at the surface. Subsequently, the chamber is purged with an inert carrier gas (typically N₂ or Ar) to remove any unreacted precursor or reaction by-products. This is then followed by the counter-reactant precursor pulse and purge, creating up to one layer of the desired material. This process is then cycled until the appropriate film thickness is achieved. Typically,

*Corresponding author: Bent, S.F. (sbent@stanford.edu)

³ authors contributed equally.

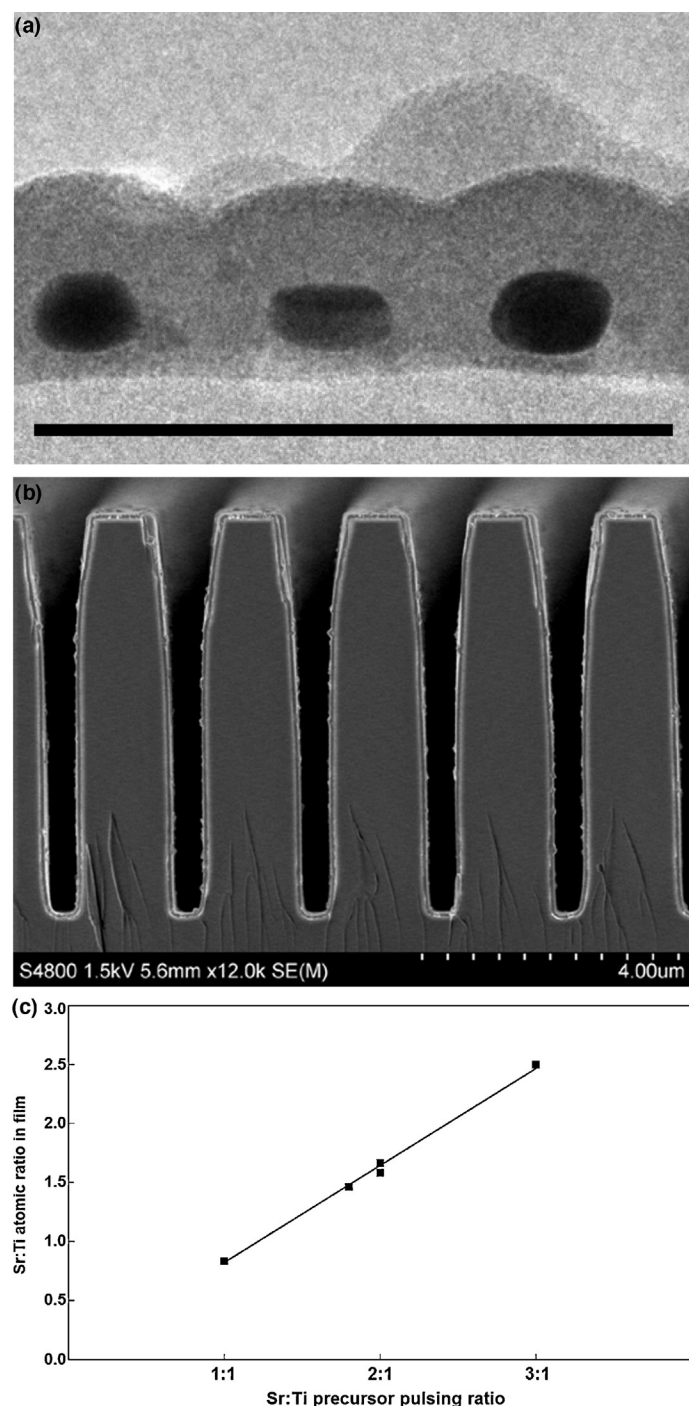
**FIGURE 1**

Schematic of ALD process. (a) Substrate surface has natural functionalization or is treated to functionalize the surface. (b) Precursor A is pulsed and reacts with surface. (c) Excess precursor and reaction by-products are purged with inert carrier gas. (d) Precursor B is pulsed and reacts with surface. (e) Excess precursor and reaction by-products are purged with inert carrier gas. (f) Steps 2–5 are repeated until the desired material thickness is achieved.

ALD processes are conducted at modest temperatures ($<350^{\circ}\text{C}$). The temperature range where the growth is saturated depends on the specific ALD process and is referred to as the ‘ALD temperature window’. Temperatures outside of the window generally result in poor growth rates and non-ALD type deposition due to effects such as slow reaction kinetics or precursor condensation (at low temperature) and thermal decomposition or rapid desorption of the precursor (at high temperature). In order to benefit from the many advantages of ALD, it is desirable to operate within the designated ALD window for each deposition process.

The primary advantages of ALD are all derived from the sequential, self-saturating, gas-surface reaction control of the deposition process. Firstly, the conformality of ALD-deposited films is often the critical factor in choosing ALD over competing deposition techniques such as CVD or sputtering. Conformality of high aspect ratio and three dimensionally-structured materials is made possible by its self-limiting characteristic, which restricts the reaction at the surface to no more than one layer of precursor. With sufficient precursor pulse times, the precursor can disperse into deep trenches, allowing for complete reaction with the entire surface. Subsequent cycles allow for uniform growth on high aspect ratio structures, whereas CVD and PVD may suffer from non-uniformity due to faster surface reactions and shadowing effects, respectively. Examples of the superior conformality of ALD films are shown in Fig. 2a and b for SnS_x films on Au nanoparticles and $\text{Ge}_2\text{Sb}_2\text{Te}_5$ films over SiO_2 trenches; the ability of the process to evenly coat

the substrate morphology is evident. A second apparent advantage of ALD is the thickness control of the deposited thin films. By utilizing layer-by-layer deposition, the thickness of a film can be tailored by the number of ALD cycles. Growth per cycle for many ALD films has been summarized in previous reviews [1,2] and is typically less than one Å/cycle, depending on the individual process. Another prominent advantage of ALD is composition control. Composition control has been demonstrated with materials such as zinc tin oxide (ZTO) [13] and SrTiO_3 [14], among many others [15–17]. These films can be deposited and compositionally controlled by tailoring ALD ‘super cycles’, which are composed of multiple ALD processes. For example, in ZTO deposition, adjusting the super cycle ratios for SnO_x and ZnO can tailor different conduction behavior and optical properties of the film [13]. In depositing SrTiO_3 , ALD processes for TiO_2 and SrCO_3 are alternated in a super cycle at a 1:1 ratio, producing a stoichiometric SrTiO_3 film after annealing (Fig. 2) [14]. However, it should be noted that a nonlinear relationship between the cycle ratio and the film’s atomic ratio is common for the ternary oxide processes [16,17], making it less straightforward to deposit a film with a certain desired composition. For example, additional studies on SrTiO_3 , ALD have indicated that a 1:1 Sr:Ti atomic ratio in films is also possible for ALD cycle ratios between 0.67 and 0.82 [15]. Other complications in composition control for ternary and quaternary metal oxide ALD arise from the need for the growth windows of individual ALD processes to be thermally compatible and the way

**FIGURE 2**

Examples of ALD strengths. (a) TEM image of conformal SnS_x ALD film on Au nanoparticles. Scale bar is 100 nm. (b) SEM cross-sectional image of conformal $\text{Ge}_2\text{Sb}_2\text{Te}_5$ ALD film in trenches [30]. (c) SrTiO_3 stoichiometry as a function of SrCO_3 and TiO_2 ALD super cycle ratio [14]. Image (b) Reprinted with permission from Ref. [30]. Copyright (2009) American Chemical Society and image (c) Reprinted (adapted) from Ref. [14], Copyright (2003), with permission from Elsevier.

in which ALD introduces dopants into the films as ‘ δ -doping’ spikes resulting in inhomogeneous films as-deposited that typically require annealing [18].

While ALD has many promising features, it also suffers from slow deposition rates. Because of the long cycle times involved in pulsing and purging precursors and the layer-by-layer nature of the

deposition, most ALD rates are on the order of 100–300 nm/h [3,12]. This rate is strongly dependent on the reactor design and the aspect ratio of the substrate, however. As the surface area and volume of an ALD reactor increase, so does the time needed for pulsing and purging. High aspect ratio substrates also require longer pulse and purge times to allow for the precursor gas to disperse into trenches and other three dimensional features. To combat this shortcoming, spatial ALD has emerged as a promising technique that may significantly improve throughput [19–23]. Spatial ALD operates by eliminating the traditional pulse/purge chamber and replacing it with a spatially-resolved head, which exposes the substrate to a specific gas precursor based on location. In one design, as the head translates around the substrate, it alters the exposed precursor, resulting in film growth. Alternatively, spatial ALD has also been demonstrated in which the substrate moves past stationary precursor nozzles, which are arranged so that by passing by them, precursor cycling is achieved and the film is grown. Overall, with spatial ALD techniques, deposition rates around 3600 nm/h are possible [24].

A wide range of materials has been grown by ALD in previous studies [1,4–10]. The materials encompass metals, insulators and semiconductors in both crystalline and amorphous phases. In addition, ALD offers a large selection of elements to choose from to create the material of choice, as listed in Table 1 [5,25–30]. This table shows that the most common types of ALD-grown materials so far are oxides, nitrides, sulfides and pure elements. Despite their more complex ALD process, compounds with three or more elements have also recently gained much interest due to their desirable materials properties and will therefore be showcased with a few examples in this review.

While the selection of materials listed in Table 1 is impressive, it clearly also shows that it is not yet possible to grow every material by ALD. The main limit of the availability of a material by ALD is the limited choice of effective reaction pathways. The selection is further limited by the availability of reactants that can facilitate the appropriate reaction pathway. As discussed earlier, it is desirable to run the ALD process in a self-limiting growth regime. This imposes a list of requirements on the reactants. Foremost, a reactant and counter reactant that will deposit the desired material must be available or synthesized. In some cases, the selection of available reactants for a given element is very limited or non-existent. The reactants should be volatile enough to be in the gas phase either at room temperature or upon moderate heating. Furthermore, while in the gas phase they should not decompose until they have reached and reacted with the sample surface. The surface reaction should preferably be quick and irreversible to lead to fast growth saturation. Neither the reactant nor its by-products, after reacting with the surface, should dissolve, etch or in other ways damage the substrate, the growing film or the ALD reactor itself. It is also preferable if the reactants are small in size to avoid reduced surface coverage due to steric hindrance. Finally, the choice of reactants has economic implications, as some are expensive and synthesis can be time consuming. Unfortunately, there are no perfect reactants, so the choice of reagents typically involves making tradeoffs between cost, availability, safety, volatility and reactivity.

There are two main groups of metal reactants: inorganic, as in the elemental and the metal halides, and metal organic, as in

TABLE 1

List of materials grown by ALD [5,25–30].

Elemental	Oxides	Nitrides	Sulfides	Other compounds
C, Al, Si, Ti, Fe, Co, Ni, Cu, Zn, Ga, Ge, Mo, Ru, Rh, Pd, Ag, Ta, W, Os, Ir, Pt	Li, Be, B, Mg, Al, Si, P, Ca, Sc, Ti, V, Cr, Mn, Fe, Co, Ni, Cu, Zn, Ga, Ge, Sr, Y, Zr, Nb, Ru, Rh, Pd, In, Sn, Sb, Ba, La, Ce, Pr, Nd, Sm, Eu, Gd, Tb, Dy, Ho, Er, Tm, Yb, Lu, Hf, W, Ir, Pt, Pb, Bi	B, Al, Si, Ti, Cu, Ga, Zr, Nb, Mo, In, Hf, Ta, W	Ca, Ti, Mn, Cu, Zn, Sr, Y, Cd, In, Sn, Sb, Ba, La, W	Li, B, Mg, Al, Si, P, Ca, Ti, Cr, Mn, Co, Cu, Zn, Ga, Ge, As, Sr, Y, Cd, In, Sb, Te, Ba, La, Pr, Nd, Lu, Hf, Ta, W, Bi

TABLE 2

Available reactant groups for specific elements [4,25–30].

Elemental	Halides	Alkyls	Cyclopentadienyls	β -diketonates	Other reactants
Mg, Mn, Zn, Ga, Cd, In, Sn	B, C, Ti, V, Cr, Mn, Cu, Zn, Ga, Ge, Zr, Nb, Mo, Cd, In, Sn, Sb, Hf, Ta, W, Pb	Be, Al, Si, Zn, Ga, Ge, Cd, In, Sn, Hg	Mg, Sc, Ni, Sr, Y, Zr, Ru, Lu, Os, Pt	Mg, Ca, Sc, V, Cr, Mn, Fe, Co, Ni, Cu, Ga, Sr, Y, Zr, Ru, Pd, In, Ba, La, Ce, Nd, Sm, Eu, Gd, Tb, Dy, Ho, Er, Tm, Ir, Pt, Pb	Li, P, Ti, Fe, Co, Ni, Cu, Zn, Ge, As, Zr, Rh, Ag, Sb, Te, La, Pr, Yb, Hf, Ta, W, Ir, Pt, Pb, Bi

alkyls, cyclopentadienyls, alkoxides, β -diketonates, amides, silyls and amidinates [4,5]. Here we describe some of the benefits and challenges associated with each type of precursor. Elemental reactants grow materials that are free of contamination and their growth is not reduced by steric hindrance, but there is only a small selection of metals that have a high enough vapor pressure to be suitable as elemental reactants. Halides are very reactive, have low steric hindrance and can be used in a wide temperature range, but suffer from having reactive by-products and from unintentional incorporation of the halide atoms into the growing material. Alkyls are also very reactive due to the direct metal carbon bond and can use H_2O as a counter reactant, but are limited by decomposition at higher temperatures and are only available for a small number of elements. Similarly, the cyclopentadienyls are reactive due to their metal carbon bonds and can use water as a counter reactant, but suffer from decomposition at higher temperatures and are only available for a small number of elements. In contrast, β -diketonates exist for a large variety of elements, but are bulky and have low reactivity leading to low growth rates. They also require more reactive counter reactants such as O_3 to achieve their highest oxide growth rates. Alkoxides suffer mainly from poor thermal stability. Amides and silyls show good reactivity, but decompose at quite low temperatures. Amidinates are a promising alternative for the future, as they are projected to be available for a big group of elements, and have decent thermal stability and good reactivity. Due to these tradeoffs and the limited choice for each element, research into new reactants is continually ongoing [5,9]. Table 2 lists the reactant groups that are currently available for a given element [4,25–30].

There is also a selection of counter reactants that are appropriately suited to the metal reactants. For instance, oxygen counter reactants include H_2O , O_3 , O_2 , H_2O_2 , and $O^\bullet$ from a plasma source [4,5]. Creating highly reactive radicals from a plasma source is a relatively new concept in ALD that is commonly denoted as plasma enhanced ALD (PEALD) [1,5–7]. While O_3 and $O^\bullet$ are very reactive and therefore can use lower deposition temperatures and enable the use of less reactive metal reactants, they are also more likely to oxidize the surface of the underlying substrate and create an unwanted interfacial layer [29–31]. Despite being less reactive,

H_2O is most commonly used for oxides, since it is gentler to the substrate surface and can withstand higher deposition temperatures without decomposing. For similar reasons, other hydrides such as NH_3 and H_2S are commonly used as counter reactants for nitrides and sulfides, respectively [4,5]. In rare cases even H_2Se is used to grow selenides, despite its toxicity [4,5].

In the following sections of this review, we will showcase some of the latest and most exciting applications of ALD. We have selected field effect transistors, thin film solar cells, and fuel cells to highlight here, because these applications each show how the unique features of ALD enable new levels of performance and deeper fundamental understanding to be achieved. These examples also illustrate the range of materials that ALD can deposit. Finally, the selection was made to show that different scientific and engineering problems can be solved by ALD, both in industry and on a research scale. In short, we hope that these applications will convince the reader that ALD is a powerful tool for solving material science problems.

ALD in microelectronics: high-k gate dielectrics

The microelectronics industry is one of the biggest adopters of ALD. Even in the late 1990s, Samsung was experimenting with ALD for improving the storage capacitor in DRAM memories [32]. In current transistor fabrication, research and development rely on using ALD for depositing pinhole-free, conformal films with well-controlled thickness and a high dielectric constant [33,34].

As industry has transitioned to the use of high-k dielectrics for the transistor gate stack in microelectronics devices, ALD has become increasingly important. The high-k gate oxides need to be highly uniform and pinhole-free on Si to prevent leakage current through the gate oxide. To solve the oxide thickness reduction challenges, Intel introduced ALD into their mass production line in 2007 [34]. This was a key reason they were able to advance from the 65 nm to the 45 nm node technology without creating transistors with significantly higher power consumption. The oxide layer they used consisted of a stack of layers: a SiON interfacial layer to electrically passivate the Si surface, a high-k HfO_2 based oxide with a k -value of around 20, and a capping layer to match the work function of the gate metal [35]. Shortly

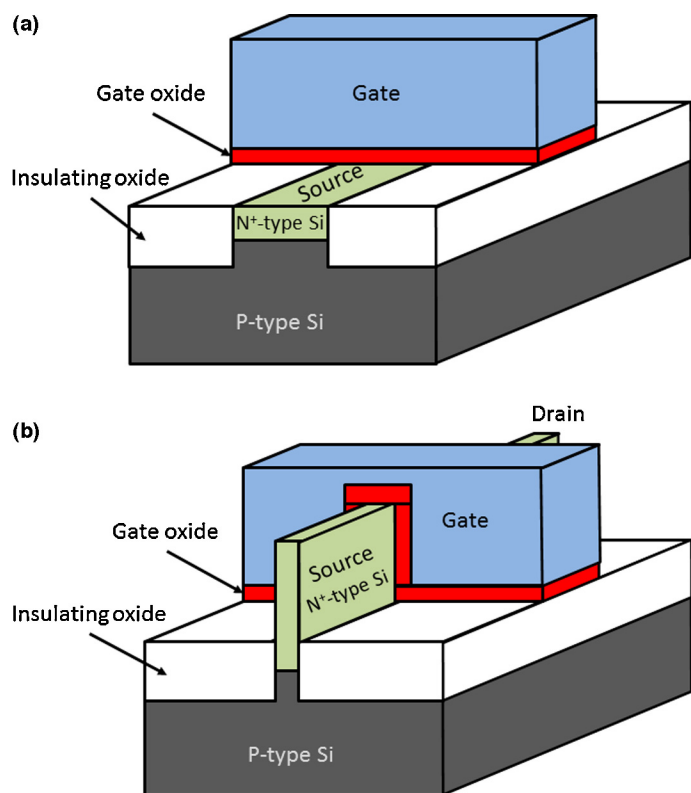


FIGURE 3

(a) The traditional planar MOSFET design leading to an inverted surface channel and (b) the FinFET or tri-gate design where a Si fin that is covered by the gate oxide from three sides becomes inverted from the surrounding gate oxide, thus increasing the overall inverted volume compared to the planar design for the same gate voltage.

afterward, other major players in the semiconductor industry followed suit and started their own production using ALD to deposit high-*k* dielectrics [36–39].

As the devices have continued to shrink, in part enabled by the ALD gate oxides which decreased the equivalent gate oxide thickness, new limitations on the bulk Si crystal have forced the industry to look for other, more radical alternatives to the traditional transistor concept. In their latest technology, the 22 nm node [33], Intel introduced a structure called tri-gate into production, which is a variant of the Fin field effect transistor (FinFET) structure. Instead of using the traditional planar channel, shown in Fig. 3a, where only the top of the Si bulk crystal is inverted, this new structure has fin like slabs of Si sticking out of the bulk crystal, as shown in Figs. 3b and 4a.

In this tri-gate design, the high aspect ratio fins protruding from the surface need to be covered with a gate oxide of great compositional and thickness uniformity and without pinholes, a task tailored for ALD. One can speculate whether such a non-planar structure would have been so readily implemented into fabrication were it not for the conformal ALD gate oxides already being a part of the existing production line.

Although information on what will be implemented by the chipmakers in the next generation technology is not public, it is informative to look toward some of the topics that are currently being researched by universities or institutes. One idea is to use the conformal ALD gate oxides and continue along the FinFET track

and further encompass the gate by undercutting it a bit on the fourth side, creating an omega gate [40], as shown in Fig. 4b. Another version is to cover the fin even below the bulk crystal surface level creating a pi gate [41] depicted in Fig. 4c. The ultimate design would be to use a semiconducting wire or tube and completely surround it by the gate [42] like the nanotube in Fig. 4d. These device structures are clearly enabled by the conformality of a process like ALD.

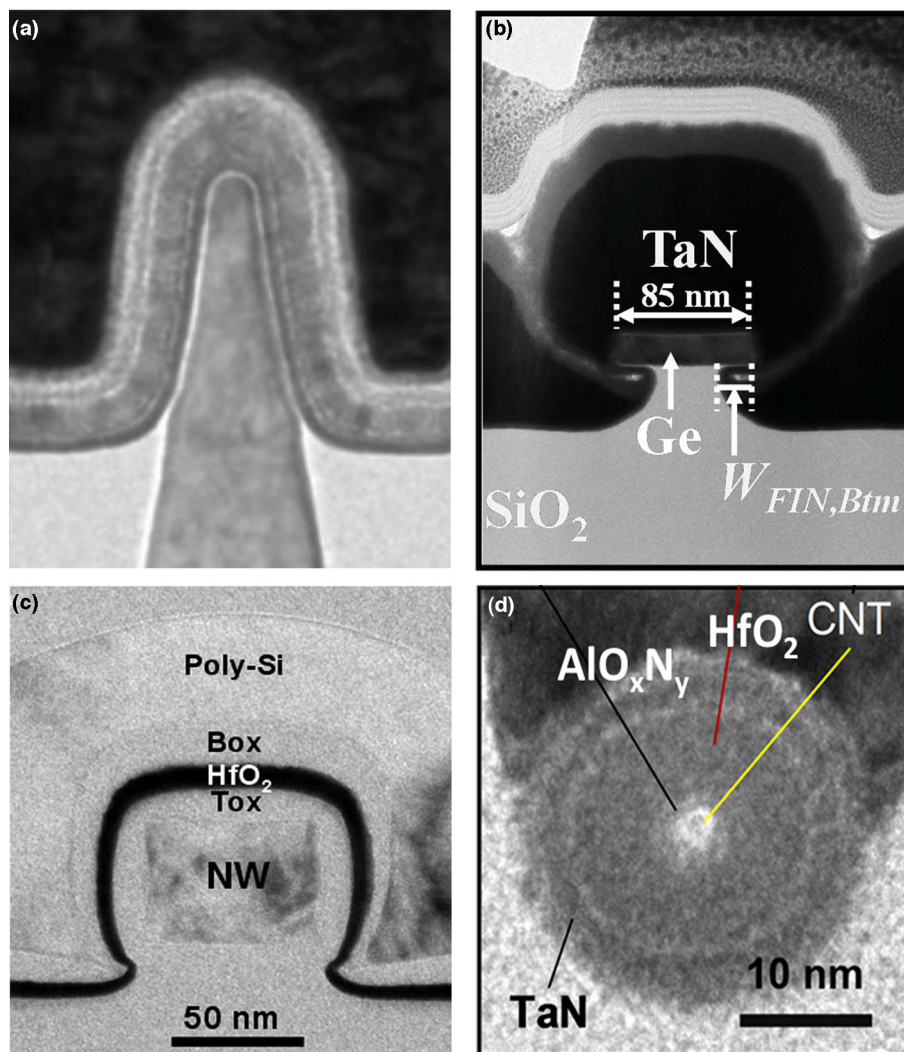
Another idea being pursued is to find a gate oxide for Si which has an even higher dielectric constant and which can be deposited by ALD. Possible candidates include SrTiO₃ [43], Al-doped TiO₂ [44], LaLuO₃ [45], Hf_{1-x}Zr_xO₂ [46], SrRuO₃ [47] and HfTiO_x [48]. Once again, ALD turns out to be an excellent tool to explore new high-*k* compounds due to its ability to grow compositionally-uniform films and its relatively straightforward way of controlling the material composition by alternating binary material cycles during growth.

Lastly, to reduce the power and to increase the speed of the devices, it is of interest to increase the electron and hole mobility within the channel for the NMOS and p-channel MOSFET (PMOS) transistors respectively. It is therefore natural to look toward alternative semiconductors to bulk silicon in which the electron or hole mobility is higher. III-V compound semiconductors are generally good examples of materials with excellent electron mobility, whereas Ge is an example of a material with an excellent hole mobility. The FinFET structure has the same advantages for these materials as well [49–51], but will require new high-*k* dielectric gate stacks that are adapted to these materials instead. Materials such as TaSiO_x, HfO₂, Al₂O₃ and HfAlO have so far successfully been implemented as high-*k* dielectrics by ALD for III-V compounds [52,53] and Al₂O₃ and HfO₂ for Ge [49,50,54]. It also is critical to form a good passivation layer on top of Ge [55] and to rid the III-V/high-*k* dielectric interface of native surface oxides that tend to pin the Fermi level at the interface [56]. Interestingly enough, certain ALD precursors as TMA have shown the ability to remove all or some of these native oxides during the ALD process [55]. This is yet another advantage that hints at ALD being an important tool for the future of electronics development.

Looking further ahead, ALD promises to be an interesting technique for applications in emerging electronics where new, lower dimensionality semiconductors using sheet, tube or wire based structures such as graphene [57], WSe₂ [58], carbon nanotubes [59] and semiconductor nanowires [60] require a conformal high-*k* gate oxide coating that does not significantly disrupt the unique low dimensional material properties. With continued scaling and adoption of new designs and materials, ALD is likely to become an even more critical tool for the microelectronics industry in the future.

ALD in photovoltaics: buffer layers in Cu(In,Ga)Se₂ thin film solar cells

There are numerous applications for ALD in photovoltaic devices [10]. Materials deposited by ALD are used as a rear contact passivation layer [61], in organic solar cells as a means to tune the electrode work function [62], and in dye sensitized solar cells as a barrier to prevent recombination [63]. Additionally, ALD can be used to create the transparent conducting oxide for the solar cell technologies listed above [64]. There is also research on creating

**FIGURE 4**

Different multiple gate design structures where ALD gate oxides have been used. (a) A TEM cross-section of Intel's [33] FinFET transistor at the 22 nm node with the gate-oxide and gate wrapped around the fin. (b) Liu *et al.* omega gate structure wrapping around a Ge channel [50]. (c) A pi-gate surrounding a poly-Si nanowire in a thin film transistor by Chen *et al.* [60]. (d) A carbon nanotube FET with a gate all around structure by Franklin *et al.* [59]. Reprinted from (a) Copyright (2012) IEEE. Reprinted, with permission, from Ref. [33], (b) Copyright (2013) IEEE. Reprinted, with permission, from Ref. [50], (c) Copyright (2013) IEEE. Reprinted, with permission, from Ref. [60], (d) Reprinted with permission from Ref. [59], Copyright (2013) American Chemical Society.

the light absorbing layer using ALD both in the form of thin films [65] and in the form of nanoparticles with quantum confinement properties [66,67]. Finally, as we will discuss in further detail in this section, it can be used in thin film solar cells to engineer the electrical junction properties of the solar cell diode.

The ability to precisely control the composition of a compound of three or more elements by ALD is very useful in optoelectronic materials such as photovoltaics as it enables a means to controllably vary properties as band gap, density, conductivity, energy band levels, and morphology. An example of this is the ALD-grown $\text{Zn}_{1-x}\text{Sn}_x\text{O}_y$ compound, where the conduction band level and valence band level were found by Kapilashrami *et al.* [68] to change as the composition is varied from ZnO to SnO_y , shown in Fig. 5.

One application where the ability to finely tune the composition and in turn the position of the conduction and valence bands is required is in $\text{Cu}(\text{In,Ga})\text{Se}_2$ (CIGS) thin film solar cells. Since CIGS is p-type as deposited, n-type layers of different semiconductors are

introduced in CIGS solar cells to form the electrical pn-junction, as shown in Fig. 6a [69]. The n-type layer deposited directly on top of the CIGS is defined as the buffer layer. Since it forms the pn-junction, the electrical properties of this buffer layer are critical. The standard material for this buffer layer is CdS, but because CdS is toxic and also parasitically absorbs blue light, much research effort has been expended on finding a good alternative. It is believed that a small positive conduction band offset (CBO) toward the CIGS is required for an optimal buffer layer [70]. Finding such a CBO is a task that is inherently well suited for an ALD process in which the material composition can be controlled. Moreover, other ALD benefits are also important for CIGS solar cells. The CIGS surface is typically very uneven, with surface roughness values in the range of 10–100 nm, and crevices can form between grains as shown in the TEM micrograph taken by Lindahl *et al.* [71] in Fig. 6b. It is therefore desirable to use a conformal buffer layer such as the ones obtained by ALD to fully cover the rough CIGS surface. This is especially important during large scale fabrication since the buffer layer must be

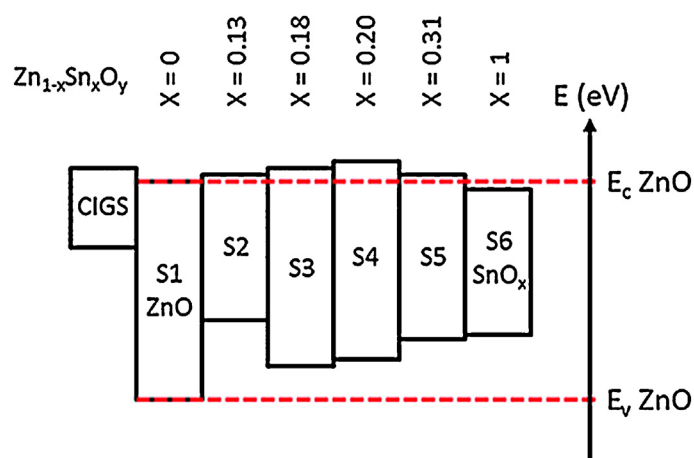


FIGURE 5

Energy band levels of ALD-grown $\text{Zn}_{1-x}\text{Sn}_x\text{O}_y$ as a function of x . The band positions were recently determined by Kapilashrami *et al.* [68] using a combination of X-ray emission spectroscopy and X-ray absorption spectroscopy. Fig. 5 is reprinted from Ref. [68], with permission from the PCCP Owner Societies.

pinhole free over the entire 1 m^2 solar cell module, in order to not severely degrade its performance.

Historically, there has been extensive research on CIGS buffer layers [72], but the binary system ZnO was the first CIGS buffer layer to be deposited by ALD [73]. The performance of the ZnO was found to be lacking, mainly due to high recombination because of the unfavorable CBO. This led to the development of ternary ALD compound systems such as $\text{Zn}(\text{O},\text{S})$ [74] and $(\text{Zn},\text{Mg})\text{O}$ [75], where the properties of ZnO could be modified so that a more favorable CBO could be achieved. Both of these ALD-grown ternary compounds were later shown to be excellent buffer layer candidates for

CIGS solar cells, each producing devices with power conversion efficiencies of more than 18%, which was comparable to or better than their corresponding reference devices using CdS. In the meantime, the ALD-grown binary compound In_2S_3 showed excellent performance creating CIGS solar cells with over 16% conversion efficiency [76]. Recently, the interest in using ALD for CIGS buffer layers has increased both for improving and understanding [76–82] the systems mentioned above and for developing new materials combinations such as $\text{Zn}_{1-x}\text{Sn}_x\text{O}_y$ [68,71,83] and ZnInS [84].

In a recent experiment by Hultqvist *et al.* [83] the composition of the $\text{Zn}_{1-x}\text{Sn}_x\text{O}_y$ buffer layer in CIGS solar cells was varied in order to find a composition where the CBO was favorable. These results further highlight the strength of ALD in carrying out compositional tuning in materials. Fig. 7 shows the resulting conversion efficiency of the solar cells as a function of the $\text{Zn}_{1-x}\text{Sn}_x\text{O}_y$ composition. From these results it is clear that there are intermediate $\text{Zn}_{1-x}\text{Sn}_x\text{O}_y$ compounds where the solar cell efficiency is higher, mainly due to a high open circuit voltage (V_{oc}), which is an indirect indicator of a small recombination current. Thus, by tuning the $\text{Zn}_{1-x}\text{Sn}_x\text{O}_y$ composition and in turn the conduction band level, a favorable CBO with reduced recombination is found for a composition of approximately $x = 0.2$. A more recent study by Lindahl *et al.* [71] showed that it was possible to push the performance of the CIGS solar cells above 18% using the right $\text{Zn}_{1-x}\text{Sn}_x\text{O}_y$ composition and thickness. This same study [71] also demonstrated that the performance of the solar cells using sufficiently thick $\text{Zn}_{1-x}\text{Sn}_x\text{O}_y$ buffer layers could be retained even if the intrinsic-ZnO layer, typically deposited to prevent shunts, was thinned or completely omitted from the CIGS solar cell stack. This reduction of a processing step, enabled by the pinhole-free, conformal coating properties of ALD, could potentially reduce the manufacturing costs of CIGS modules.

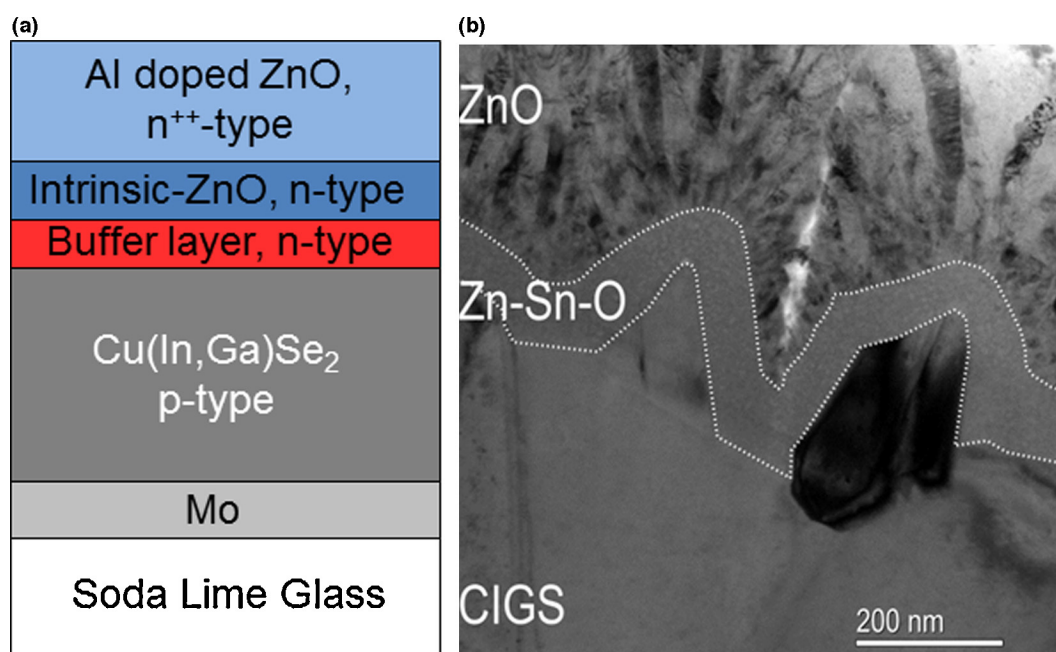
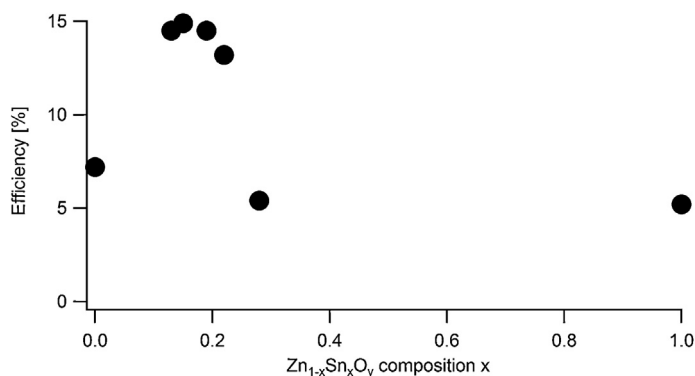


FIGURE 6

(a) Conceptual CIGS solar cell stack and (b) the corresponding TEM cross section, where the CIGS layer has a conformal coating of ALD $\text{Zn}_{1-x}\text{Sn}_x\text{O}_y$ [71]. To more easily distinguish between the layers in the TEM micrograph, a dotted line has been added to define the CIGS/ $\text{Zn}_{1-x}\text{Sn}_x\text{O}_y$ and $\text{Zn}_{1-x}\text{Sn}_x\text{O}_y$ /intrinsic ZnO interfaces. Fig. 6 is reprinted from Ref. [71], Copyright (2012) John Wiley & Sons Ltd.

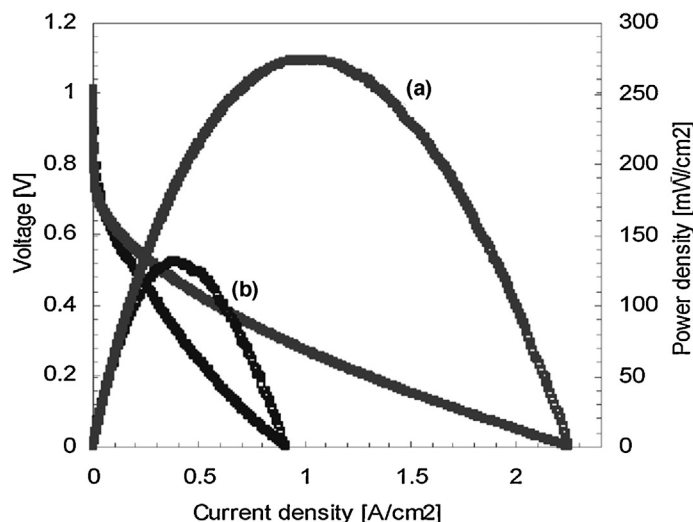
**FIGURE 7**

Measured solar cell conversion efficiency as a function of the $\text{Zn}_{1-x}\text{Sn}_x\text{O}_y$ composition. Adapted with permission from Ref. [83], Copyright (2012) John Wiley & Sons Ltd.

In general, ALD-grown buffer layers for CIGS solar cells have been shown in multiple reports to give comparable conversion efficiencies as CdS. While these ALD layers lead to higher J_{sc} , they rarely outperform traditional CdS in terms of V_{oc} , an effect attributed to higher interface recombination with the alternative buffer layers. The search therefore continues to find compounds that can match or outperform CdS in terms of V_{oc} . In this search, ALD has become a very important tool due to its ability to control material properties of ever more complex compounds. Moreover, the ability to tune the electronic properties of the buffer layer has also spurred interest for ALD among groups currently developing novel crystalline thin film absorber materials as $\text{Cu}_2(\text{Zn},\text{Sn})(\text{S},\text{Se})_4$ [85], SnS [86] and Cu_2O [87], where similar challenges arise as in the CIGS case.

ALD in energy storage: Pt and YSZ for solid oxide fuel cells

Another application in which ALD shows good potential is in next-generation fuel cells, particularly solid oxide fuel cells (SOFCs). SOFCs are solid-state electrochemical devices capable of converting chemical energy, in the form of hydrogen or other hydrocarbon gases, into electrical energy. SOFCs generally consist of three layers – a porous anode material, an ion conducting electrolyte, and a cathode material, all of which can utilize the benefits of ALD. Operating temperatures for typical SOFC devices are 800–1000 °C. Much of the current research into SOFCs is focused on reducing the operating temperatures to ranges between 300 and 600 °C [88–96]. To mitigate limitations associated with lower temperatures, such as lower ionic conductivity and slower reaction kinetics at the electrodes, different strategies have been employed. One approach is to reduce the resistance of the electrolyte, either through the synthesis of a material that has a higher ionic conductivity at lower temperature ranges (300–600 °C) or by reducing the electrolyte film thickness [88,89,92–96]. The latter method offers a two-fold advantage: besides the lower resistance of the device, the thin film allows for rapid ion transfer between electrodes due to the decreased diffusion length [97]. Another approach is to enhance the reaction kinetics at the electrodes with catalysts that are effective at lower temperatures. Both of these approaches can benefit from the advantages imparted by ALD, and a number

**FIGURE 8**

Polarization curves and power density of YSZ fuel cells. Performance comparison of (a) ALD YSZ fuel cells to (b) reference fuel cells with 50 nm RF-sputtered YSZ at 350 °C [92]. Adapted with permission from Ref. [92], Copyright (2007) American Chemical Society.

of studies have applied ALD to deposit both electrolytes and catalysts.

The electrolyte in SOFCs is generally yttria stabilized zirconia (YSZ) or gadolinia-doped ceria (GDC), which are dense, ion-conducting ceramics. Several studies have investigated growth of YSZ by ALD, with focus on both improving ionic conductivity and significantly reducing the electrolyte film thickness. YSZ electrolyte has been deposited by alternating ZrO_2 and Y_2O_3 ALD depositions to reach the desired YSZ composition [92,94,98]. In a study by Shim *et al.*, YSZ thin films prepared by ALD were compared to reference films of YSZ prepared by radio frequency (RF) sputtering. $\text{ZrO}_2:\text{Y}_2\text{O}_3$ composition ratios of the YSZ were at 0.92:0.08 (8 mol% Y dopant), a value often targeted for the bulk composition of YSZ for optimal results in SOFCs [94,98,99]. Electrodes for both ALD and RF-sputtered SOFCs were 80 nm of porous Pt. It was demonstrated that 60 nm ALD YSZ films had a power density of 270 mW/cm^2 and activation energy of 0.88 eV, while 50 nm RF sputtered YSZ films had a power density of only 130 mW/cm^2 at 350 °C, as shown in Fig. 8. Additionally, Shim noted that the exchange current density of the ALD-YSZ/Pt interface was four orders of magnitude greater than reference YSZ/Pt interfaces, suggesting the nanocrystalline morphology of the YSZ ALD films enhances the catalytic properties of oxygen dissociation [92].

More recently, investigations of YSZ thin films deposited by ALD have looked at the composition of the YSZ electrolyte and its effect on ion conductivity and catalytic activity. Son *et al.* investigated the ionic conductivity of ALD-grown YSZ, varying the $\text{ZrO}_2:\text{Y}_2\text{O}_3$ ALD cycle ratios from 1:1 to 6:1 [94]. These cycle ratios and their respective growth rates correlated to a dopant mole ratio ranging from 28.8 to 4.3 mol%. The 4:1 cycle ratio (10.4 mol% Y dopant) resulted in the highest conductivity of all films, which was 2 orders of magnitude greater than bulk YSZ (8 mol% Y dopant). This large difference was attributed to the inhomogeneity of the dopant in the vertical direction of the device that arises from the

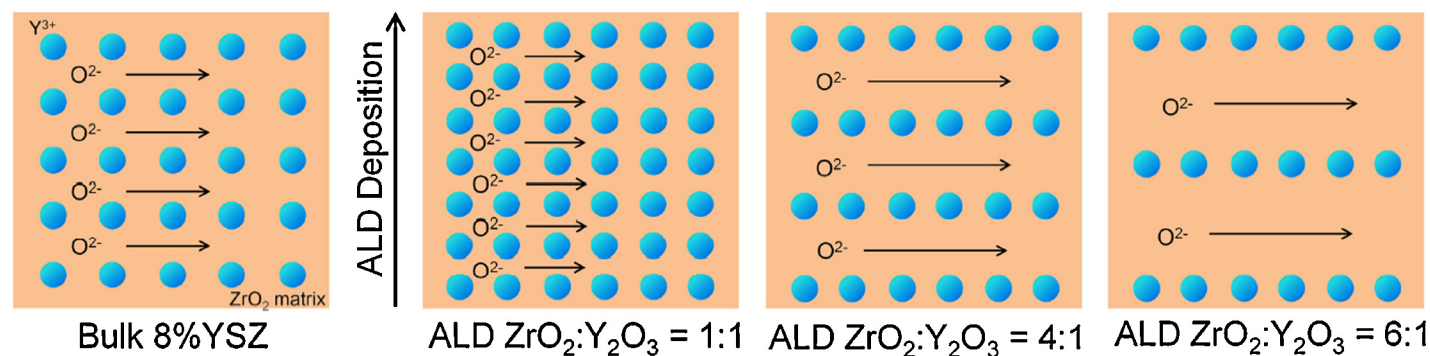


FIGURE 9

Conceptual illustration of dopant ion positioning in bulk YSZ and ALD YSZ films [94]. Reprinted (adapted) with permission from Ref. [94], Copyright (2009) American Vacuum Society.

cyclic nature of the ALD process. This is demonstrated in Fig. 9, where yttria dopants (blue dots) vary spatially in the vertical direction, depending on the ALD cycle ratio. Theoretical evaluation of the spatial doping and its relation to ionic conductivity are currently being investigated. In a related study by Chao *et al.* [98], a surface modification layer (7 nm) of YSZ deposited onto bulk YSZ (500 μm) was investigated for its surface exchange and transport properties. Films deposited with a 2:6 ratio of Y₂O₃:ZrO₂ ALD cycles (14 mol% Y dopant) resulted in a fivefold increase in the surface exchange coefficient over bulk YSZ. However, none of the films exhibited improved transport properties.

In addition to electrolyte studies using ALD, much effort has been focused on catalyzing the oxidation and reduction reactions at the electrodes. With reaction activation dropping with operating temperature, many strategies to enhance the reaction kinetics have been investigated. A popular method has been through sputtered and ALD deposited Pt metal for the cathode [90,91]. Pt can be deposited via ALD by (methylcyclopentadienyl) trimethylplatinum (MeCpPt-Me₃) precursor and an air (oxygen) counter reactant. Pt-ALD is typically performed around 275–300 °C. Pt-ALD and noble metal ALD, in general, suffer from slow nucleation rates. However, one of the advantages to using ALD to deposit Pt over sputtering is the intrinsic deposition roughness of noble metal ALD processes, which increases the surface area (triple phase boundary) of the catalyst, leading to a more efficient device [91].

An early study from Jiang *et al.* of Pt electrode catalysts highlighted the opportunity for Pt thin films in SOFCs [90]. In their investigation, the thickness of ALD-grown Pt was varied between 5 nm, 10 nm, and 30 nm and compared to films of 80 nm of Pt deposited via direct current (dc)-sputtering. With the ALD films, maximum catalytic electrode performance was found with 10 nm thick Pt at 50 mW/cm², but 80 nm dc-sputtered Pt films had a higher power density of 80 mW/cm² at an operating temperature of 350 °C. This is likely due to the differences in porosity of the two deposition methods. The 10 nm ALD film had porosities in the range of 15–20% while the dc-sputtered films had porosities between 30 and 40%, allowing better fuel diffusivity. Significantly, their study showed that ALD fabrication reduced the Pt loading needed in the SOFC. More recently, An *et al.* [91] investigated the triple phase boundary density of ALD films as a function of ALD cycles and their resulting power density. These ALD thin films were also compared to 80 nm dc-sputtered Pt films. Their results were

consistent with those of Jiang *et al.* and showed that while device improvement was not significant, Pt loading was significantly decreased. The ALD Pt films only required roughly 10% of the Pt metal that dc-sputtered films required for similar device performance. This reduced Pt loading originated from the contiguity and the high triple phase boundary density, as TEM analysis indicated. The reduced loading is a critical finding for the long term economical prospects of using Pt catalysts for microscale SOFCs.

In another study, Jiang *et al.* [100] utilized area-selective ALD processes to create Pt current collector grids on a La_{0.6}Sr_{0.4}Co_{0.2}Fe_{0.8}O₃ cathode. To generate the grid pattern, microcontact printing methods were utilized, along with an octadecyltrichlorosilane (ODTS) monolayer. ODTS has been shown in previous studies to deactivate the surface for Pt ALD processes [101]. This allowed the Pt metal to be patterned into a grid structure. Power densities of μ -SOFC with and without the Pt grid were 110.7 and 11.2 $\mu\text{W}/\text{cm}^2$, respectively. This order of magnitude increase is credited to the enhanced oxygen dissociative adsorption onto the cathode from the Pt metal along with the collection of more current from the cathode.

In summary, ALD techniques have shown great promise in matching performance of sputtered SOFC devices, with the advantage of reducing material loading required for a specific performance. As more is understood about the structure and triple phase boundary of the cathode catalysts and ion conducting electrolytes, ALD, with its excellent composition and thickness control, has the potential to play a pivotal role in the development of next generation SOFCs.

Summary/outlook

With devices becoming ever smaller and increasingly structured into complex three dimensional shapes, the need for controllable and conformal thin films has never been greater. ALD, with its sequential self-limiting reactions, is able to meet these demands in one of the most effective methods possible. Comparable techniques, such as CVD and PVD, cannot always provide the same level of uniformity, conformality and thickness control at the Angstrom level. Because of the advantages of ALD, ALD processes have been developed for a wide variety of materials, ranging from metals to metal oxides to complex ternary materials, allowing ALD to become incorporated into industrial procedures.

In this review, we briefly highlighted many of the advantages of ALD, such as conformality, thickness control, and composition control, along with a broad range of applications in which ALD can be utilized. These applications encompass different technologies, such as the energy conversion industry (photovoltaics, fuel cells) and the semiconductor industry (high-k transistors). In CIGS solar cell devices, ALD has been used to deposit conformal thin films of $\text{Zn}_{1-x}\text{Sn}_x\text{O}_y$ buffer layers. By tailoring the composition and thickness of the $\text{Zn}_{1-x}\text{Sn}_x\text{O}_y$ films via ALD, the device efficiency could be maximized at 18% for an approximate value of $x = 0.20$. For SOFCs, ALD has been utilized in Pt catalyst deposition, showing comparable power densities of devices with dc-sputtered Pt, but requiring only 10% of the Pt metal thickness. Additionally, ALD has been used to deposit thin films of YSZ electrolyte in an effort to reduce the ion exchange distance. By manipulating the $\text{Y}_2\text{O}_3:\text{ZrO}_2$ ratio in ALD super cycles, the composition of the resulting YSZ films can be controlled to provide superior surface exchange and transport properties. Finally, this review showcased emerging high-k dielectric materials, for which ALD has been used to deposit new materials such as Al doped TiO_2 , LaLuO_3 , $\text{Hf}_{1-x}\text{Zr}_x\text{O}_2$, SrRuO_3 , and HfTiO_x . Also, as transistors have evolved, traditional high-k dielectric ALD procedures, such as that for HfO_2 , have been incorporated for use with FinFETs, providing a uniform and conformal coating on the fin.

As these devices continue to develop, a general emerging trend is the shift from traditional two dimensional planar devices to highly ordered three dimensional structures, often to maximize surface areas and densities to improve the device efficiency. As researchers push the envelope for these smaller complex structures, the demand for precise deposition techniques has never been higher, leading the way for ALD to play an integral role in further device development. With newer technologies like spatial ALD, which allows for faster deposition speeds, and commercially available ALD reactors, the presence of ALD in manufacturing will continue to increase as more materials systems are developed. As the number of ALD systems grows, look for ALD to expand not only in the industries reviewed in this article, but into newer, novel applications such as the medical industry, used for example, in functionalized pharmaceuticals or implanted detectors, and tribology, used for example to create super durable coatings on various surfaces, as well as many others that can benefit from its unique and precise control over thickness, composition, and conformality.

Acknowledgments

This work was supported by the U.S. Department of Energy, Office of Science, Basic Energy Sciences, under Award # DE-SC0004782. The authors also acknowledge fellowship support by the J. Hewes Crispin and Marjorie Holmes Crispin Stanford Graduate Fellowship (RWJ) and the Marcus and Amalia Wallenberg Foundation under the Stig Hagstöm Stipend (AH).

References

- [1] S.M. George, *Chem. Rev.* 110 (1) (2009) 111–131.
- [2] M. Knez, K. Nielsch, L. Niinistö, *Adv. Mater.* 19 (21) (2007) 3425–3438.
- [3] M. Leskelä, M. Ritala, *Angew. Chem. Int. Ed.* 42 (45) (2003) 5548–5554.
- [4] R.L. Puurunen, *J. Appl. Phys.* 97 (12) (2005), 121301–121301-52.
- [5] V. Miikkulainen, et al. *J. Appl. Phys.* 113 (2) (2013), 021301–1–021301-101.
- [6] H. Kim, *J. Vac. Sci. Technol. B: Microelectron. Nanometer Struct.* 21 (6) (2003) 2231–2261.
- [7] M. Leskelä, M. Ritala, *Thin Solid Films* 409 (1) (2002) 138–146.
- [8] J.A. An Delft, D. Garcia-Alonso, W.M.M. Kessels, *Semicond. Sci. Technol.* 27 (7) (2012) 074002.
- [9] T.J. Knisley, L.C. Kalutarage, C.H. Winter, *Coord. Chem. Rev.* 257 (23–24) (2013) 3222–3231.
- [10] J.R. Bakke, et al. *Nanoscale* 3 (9) (2011) 3482–3508.
- [11] T. Suntola, J. Antson, inventors, Method for producing compound thin films, Patent 4058430, November 15, 1977.
- [12] M. Ritala, J. Niinistö, *Chemical Vapour Deposition: Precursors, Processes and Applications*, Royal Society of Chemistry, 2009.
- [13] A. Hultqvist, M. Edoff, T. Törndahl, *Prog. Photovoltaics Res. Appl.* 19 (4) (2011) 478–481.
- [14] A. Kosola, et al. *Appl. Surf. Sci.* 211 (1) (2003) 102–112.
- [15] M. Vehkamäki, et al. *Electrochem. Solid-state Lett.* 2 (10) (1999) 504–506.
- [16] S.D. Elliott, O. Nilsen, *ECS Trans.* 41 (2) (2011) 175–183.
- [17] H. Wang, et al. *Electrochem. Solid-State Lett.* 12 (4) (2009) G13–G15.
- [18] A. Van Gorkum, K. Nakagawa, Y. Shiraki, *J. Appl. Phys.* 65 (6) (1989) 2485–2492.
- [19] F. Werner, et al. *Appl. Phys. Lett.* 97 (2010) 162103.
- [20] P. Poodt, et al. *J. Vac. Sci. Technol. A* 30 (1) (2012), 010802-010802-11.
- [21] D.H. Levy, S.F. Nelson, D. Freeman, *J. Display Technol.* 5 (12) (2009) 484–494.
- [22] D.H. Levy, et al. *Appl. Phys. Lett.* 92 (19) (2008), 192101-192101-3.
- [23] P. Poodt, et al. *Adv. Mater.* 22 (32) (2010) 3564–3567.
- [24] P. Poodt, et al. *J. Electrochem. Soc.* 158 (9) (2011) H937–H940.
- [25] J. Hämäläinen, et al. *Chem. Mater.* 24 (1) (2011) 55–60.
- [26] J. Yum, et al. *J. Appl. Phys.* 109 (6) (2011), 064101-064101-4.
- [27] K. Uusi-Esko, M. Karppinen, *Chem. Mater.* 23 (7) (2011) 1835–1840.
- [28] I. Povey, et al. *Appl. Phys. Lett.* 89 (10) (2006), 104103-1–104103-3.
- [29] H.B. Park, et al. *J. Appl. Phys.* 94 (2003) 3641.
- [30] V. Pore, et al. *J. Am. Chem. Soc.* 131 (10) (2009) 3478–3480.
- [31] S. Ha, E. Choi, S. Kim, *Thin Solid Films* 476 (2) (2005) 252–257.
- [32] J.S. Lim, et al. *ICVC '99 6th International Conference*, 1999, 506–509.
- [33] C. Auth, et al. *2012 Symposium on VLSI Technology (VLSIT)*, 2012, 131–132.
- [34] K. Mistry, et al. *Electron Devices Meeting 2007, IEDM 2007, IEEE International*, 2007, pp. 247–250.
- [35] K.J. Kuhn, *IEEE Trans. Electron Dev.* 59 (7) (2012) 1813–1828.
- [36] S. Hyun, et al. *2011 Symposium on VLSI Technology (VLSIT)*, 2011, 32–33.
- [37] S. Narasimha, et al. *Electron Devices Meeting (IEDM)*, 2012 IEEE International, 2012, 3.3.1–3.3.4.
- [38] F. Koehler, et al. *IOP Conference Series: Materials Science and Engineering*, vol. 41(1), 2012, . p. 012006.
- [39] C.H. Tsai, et al. *2012 International Symposium on VLSI Technology Systems and Applications (VLSI-TSA)*, 2012, 1–2.
- [40] F.-L. Yang, et al. *Electron Devices Meeting, 2002 IEDM '02 International*, 2002, 255–258.
- [41] J.-T. Park, J. Colinge, C.H. Diaz, *IEEE Electron Dev. Lett.* 22 (8) (2001) 405–406.
- [42] S. Bangsaruntip, et al. *2010 Symposium on VLSI Technology (VLSIT)*, 2010, 21–22.
- [43] V. Elshocht, *Electron Devices Meeting 2008, IEDM 2008, IEEE International*, 2008, 1–4.
- [44] Z.-Y. Xie, et al. *2012 IEEE 11th International Conference on Solid-State and Integrated Circuit Technology (ICSICT)*, 2012, 1–3.
- [45] S. Shen, et al. *Appl. Phys. Lett.* 98 (17) (2011), 172902-1–172902-3.
- [46] C.K. Chiang, et al. *Reliability Physics Symposium (IRPS)*, 2012 IEEE International, 2012, GD.3.1–GD.3.4.
- [47] J. Hwan Han, et al. *Semiconductor Device Research Symposium (ISDRS)*, 2011 International, 2011, 1–2.
- [48] K. Tomida, et al. *IOP Conference Series: Materials Science and Engineering*, vol. 8(1), 2010, . p. 012023.
- [49] C.-W. Chen, et al. *IEEE Electron Dev. Lett.* 33 (12) (2012) 1678–1680.
- [50] B. Liu, et al. *IEEE Trans. Electron Dev.* 60 (6) (2013) 1852–1860.
- [51] M. Radosavljevic, et al. *Electron Devices Meeting (IEDM)*, 2010 IEEE International, 2010, 6.1.1–6.1.4.
- [52] P.D. Ye, *J. Vac. Sci. Technol. A* 26 (4) (2008) 697–704.
- [53] M. Radosavljevic, et al. *Electron Devices Meeting (IEDM)*, 2009 IEEE International, 2009, 1–4.
- [54] S. Siddiqui, et al. *ECS Trans.* 53 (3) (2013) 137–146.
- [55] R.M. Wallace, et al. *MRS Bull.* 34 (7) (2009) 493.
- [56] W. Wang, et al. *Microelectron. Eng.* 88 (7) (2011) 1061–1065.
- [57] B. Fallahazad, et al. *Appl. Phys. Lett.* 100 (9) (2012) 093112.
- [58] H. Fang, et al. *Nano Lett.* 12 (7) (2012) 3788–3792.
- [59] A.D. Franklin, et al. *Nano Lett.* 13 (6) (2013) 2490–2495.

- [60] L.-J. Chen, et al. *IEEE Trans. Nanotechnol.* 10 (2) (2011) 260–265.
- [61] J. Benick, et al. *Appl. Phys. Lett.* 92 (25) (2008), 253504-1–253504-3.
- [62] H. Schmidt, et al. *Appl. Phys. Lett.* 96 (24) (2010), 243305-1–243305-3.
- [63] E. Palomares, et al. *J. Am. Chem. Soc.* 125 (2) (2003) 475–482.
- [64] K. An, et al. *J. Nanosci. Nanotechnol.* 8 (9) (2008) 4856–4859.
- [65] A.B.F. Martinson, et al. *Appl. Phys. Lett.* 94 (12) (2009), 123107-1–123107-3.
- [66] N.P. Dasgupta, et al. *Nano Lett.* 11 (3) (2011) 934–940.
- [67] T.P. Brennan, et al. *Adv. Energy Mater.* 1 (6) (2011) 1169–1175.
- [68] M. Kapilashrami, et al. *Phys. Chem. Chem. Phys.* 14 (29) (2012) 10154–10159.
- [69] S. Niki, et al. *Prog. Photovoltaics Res. Appl.* 18 (6) (2010) 453–466.
- [70] A. Niemegeers, M. Burgelman, A. De Vos, *Appl. Phys. Lett.* 67 (6) (1995) 843–845.
- [71] J. Lindahl, et al. *Prog. Photovoltaics Res. Appl.* 21 (8) (2013) 1588–1597.
- [72] N. Naghavi, et al. *Prog. Photovoltaics Res. Appl.* 18 (6) (2010) 411–433.
- [73] V. Lujala, et al. *Appl. Surf. Sci.* 82–83 (0) (1994) 34–40.
- [74] E.B. Yousfi, et al. *Thin Solid Films* 361–362 (0) (2000) 183–186.
- [75] T. Törndahl, et al. *Prog. Photovoltaics Res. Appl.* 15 (3) (2007) 225–235.
- [76] N. Naghavi, et al. *Prog. Photovoltaics Res. Appl.* 11 (7) (2003) 437–443.
- [77] K. Ramanathan, et al., Comparative Study of Zn(O,S) Buffer Layers and CIGS Solar Cells Fabricated By CBD, ALD, and Sputtering, (2012) NREL Report No. CP-5200-54138.
- [78] T. Nakada, et al. *IEEE J. Photovoltaics* 3 (1) (2013) 461–466.
- [79] N. Naghavi, et al. *Thin Solid Films* 519 (21) (2011) 7600–7605.
- [80] K. Nakashima, et al. *Jpn. J. Appl. Phys.* 51 (2012), 10NC15.
- [81] C. Lee, Y.M. Shin, B.T. Ahn, *ECS Trans.* 41 (4) (2011) 213–219.
- [82] J. Rousset, et al. *Sol. Energy Mater. Sol. Cells* 95 (6) (2011) 1544–1549.
- [83] A. Hultqvist, et al. *Prog. Photovoltaics Res. Appl.* 20 (7) (2012) 883–891.
- [84] P. Genevee, et al. *J. Vac. Sci. Technol. A* 31 (1) (2013) 01A131.
- [85] D.A.R. Barkhouse, et al. *Appl. Phys. Lett.* 100 (19) (2012), 193904-1–193904-5.
- [86] P. Sinsermsuksakul, et al. *Appl. Phys. Lett.* 102 (5) (2013), 053901-1–053901-5.
- [87] Y. Lee, et al. *Energy Environ. Sci.* 6 (2013) 2112–2118.
- [88] D. Beckel, et al. *J. Power Sources* 173 (1) (2007) 325–345.
- [89] H. Huang, et al. *J. Electrochem. Soc.* 154 (1) (2007) B20–B24.
- [90] X. Jiang, et al. *Chem. Mater.* 20 (12) (2008) 3897–3905.
- [91] J. An, Y. Kim, F.B. Prinz, *Phys. Chem. Chem. Phys.* 15 (20) (2013) 7520–7525.
- [92] J.H. Shim, et al. *Chem. Mater.* 19 (15) (2007) 3850–3854.
- [93] C. Brahim, et al. *Appl. Surf. Sci.* 253 (8) (2007) 3962–3968.
- [94] K. Sik Son, et al. *J. Vac. Sci. Technol. A* 31 (1) (2013), 01A107-1–01A107-4.
- [95] J.H. Shim, et al. *Chem. Mater.* 21 (14) (2009) 3290–3296.
- [96] C. Bernay, et al. *J. Phys. Chem. Solids* 64 (9) (2003) 1761–1770.
- [97] A. Evans, et al. *J. Power Sources* 194 (1) (2009) 119–129.
- [98] C. Chao, et al. *ACS Nano* 7 (3) (2013) 2186–2191.
- [99] B. Steele, *Solid State Ionics* 75 (1995) 157–165.
- [100] X. Jiang, R. Chen, S.F. Bent, *Surf. Coat. Technol.* 201 (22) (2007) 8799–8807.
- [101] R. Chen, S.F. Bent, *Adv. Mater.* 18 (8) (2006) 1086–1090.